Phys 153: Fundamentals of Physics III

Unit #7 – Quantum Mechanics

An Introduction to Wave Mechanics

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Spring 2027

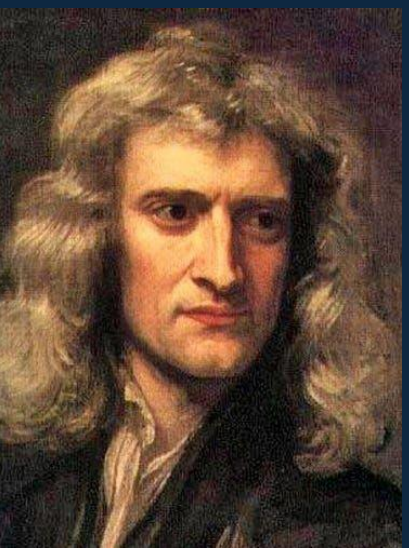
Classical vs Quantum

Classical Mechanics

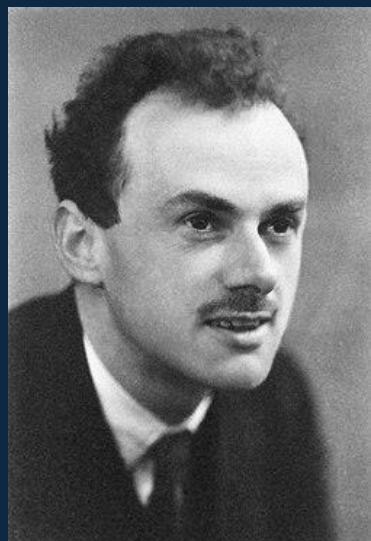
- Particle has position and velocity
- Motion determined by Newton's laws
- Measurements reveal pre-existing values

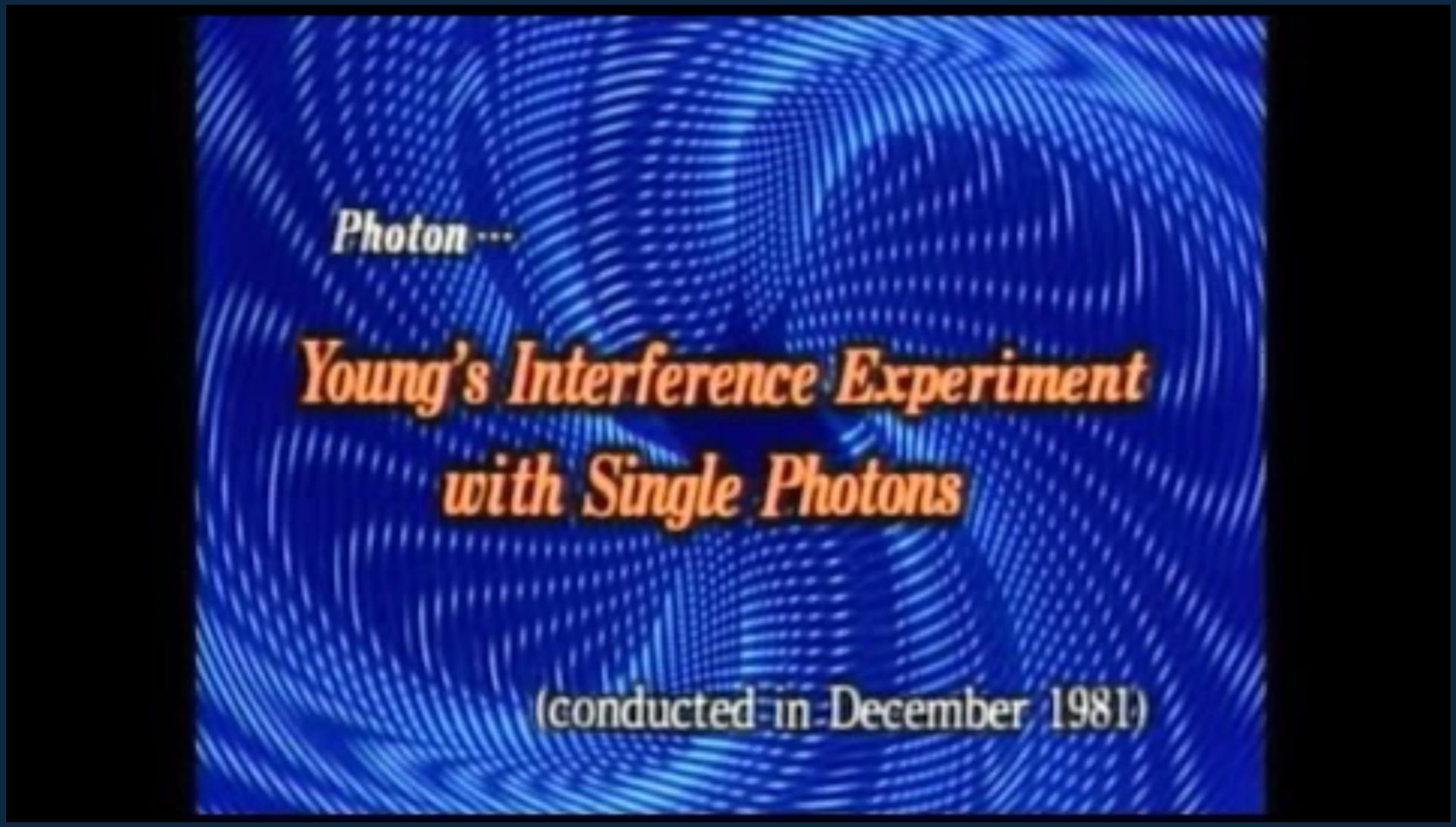
Quantum Mechanics

- Particle described by wave function $\psi(x,t)$
- Evolution of the wave function is determined by the Schrödinger equation
- Measurement outcomes are probabilistic; statistical distributions are encoded by $\psi(x,t)$








Photon ···
Young's Interference Experiment
with Single Photons
(conducted in December 1981)

The Wave Function

- $\psi(x,t)$ is also called the probability amplitude of the particle
- $|\psi(x,t)|^2$ is the probability density of finding the particle near position x (probability per unit length)
- $|\psi(x,t)|^2 dx$ is the probability of finding the particle near x within a very small interval dx
- The probability of finding the particle in the range $a < x < b$ is

$$P(a < x < b) = \int_a^b |\psi(x,t)|^2 \, dx$$

- The interference patterns seen in experiments arise from how the probability amplitudes combine—analogous to wave behavior, hence the name “wave function.”

Normalization

The square-modulus of the wave function represents a probability distribution, and it must be normalized, i.e., the total probability to find the particle *somewhere* in the range $-\infty < x < \infty$ must be one.

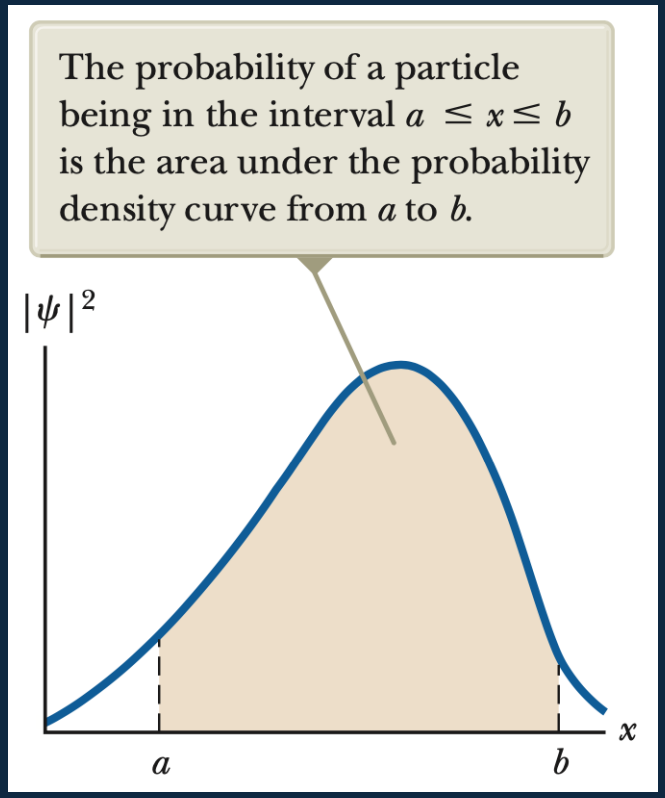
$$\int_{-\infty}^{\infty} |\psi(x,t)|^2 dx = 1$$

Normalization constitutes a constraint on the wave function.

Any physically meaningful wave function must solve the Schrödinger equation and satisfy normalization.

$\psi(x,t)$ must go to zero as $x \to \pm\infty$, otherwise it could not be normalized.

Expectation Values



Average position

$$\langle x \rangle = \int_{-\infty}^{\infty} x \, |\psi|^2 \, dx$$

Expectation value

$$\langle f(x) \rangle = \int_{-\infty}^{\infty} f(x) \, |\psi|^2 \, dx$$

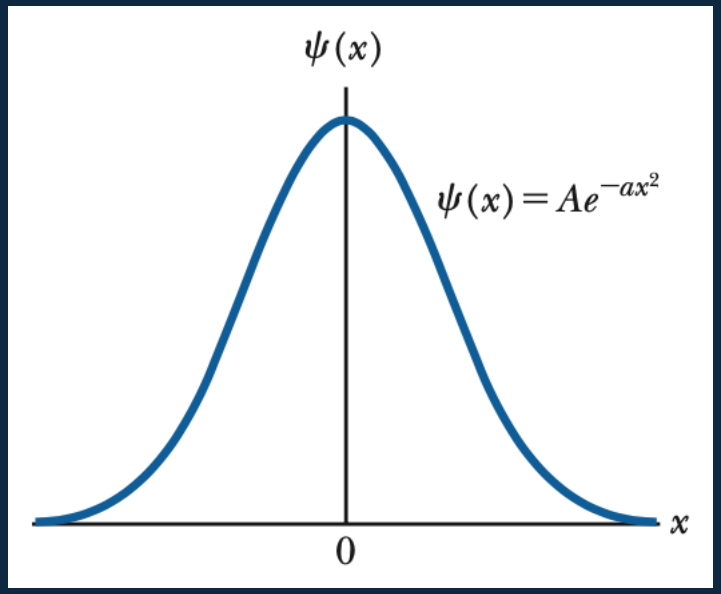
An expectation value is a weighted average, weighted by the probability

Example: Wave Function

Consider a particle whose wave function is graphed in the figure and is given by $\psi(x) = Ae^{-ax^2}$ where $A > 0$.

(a) What is the value of A if this wave function is normalized?

(b) What is the average position for this particle?

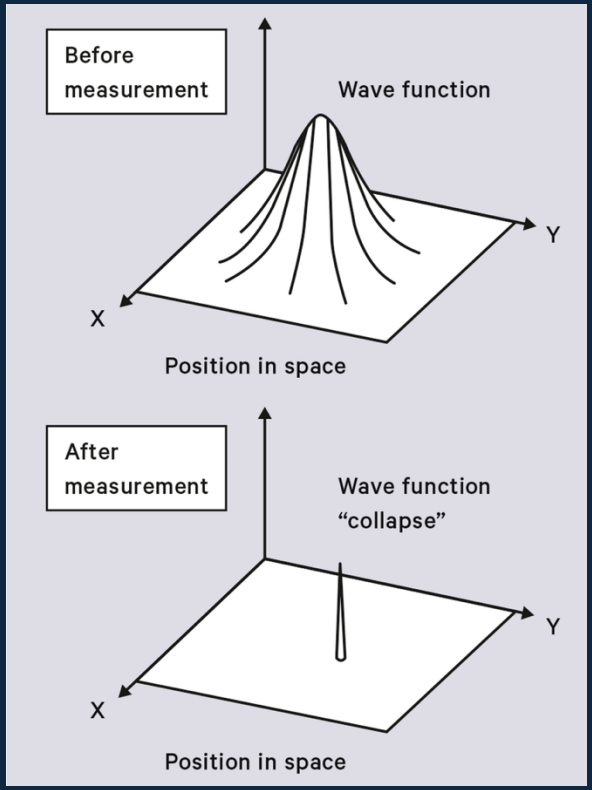


$$\int_{-\infty}^{\infty} e^{-x^2}\,dx = \sqrt{\pi}$$

Wave Function Collapse

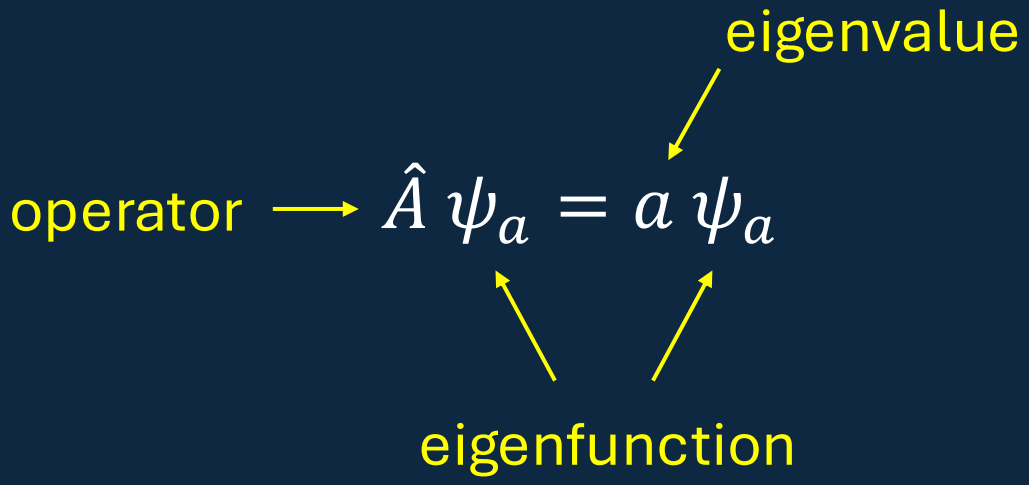
A particle is a localized object. When its position is measured, the wave function "collapses."

- No one completely understands this process, and many doubt that it's even physical.
- The wave function is primarily a tool for calculation—this does not undermine its importance!



Observables

- In quantum mechanics, quantities such as position, momentum, and energy are called observables and are represented by operators that "act on" the wave function.
- When we perform a measurement of some quantity, the only possible results that can be obtained are the eigenvalues of the corresponding operator.
- The allowed measurement outcomes are determined by solving an eigenvalue equation:



Energy & Momentum

Plane wave: $\psi(x,t) = Ae^{i(kx-\omega t)}$

$$E = hf = \frac{h}{2\pi}\, 2\pi f = \hbar\omega$$

$$p = \frac{h}{\lambda} = \frac{h}{2\pi}\frac{2\pi}{\lambda} = \hbar k$$

$$\frac{\partial}{\partial t}\psi = -i\omega\,\psi = -i\frac{E}{\hbar}\,\psi$$

$$\frac{\partial}{\partial x}\psi = ik\,\psi = i\frac{p}{\hbar}\,\psi$$

$$\hat{H}\,\psi = i\hbar\frac{\partial}{\partial t}\psi = E\,\psi$$

$$\hat{p}\,\psi = -i\hbar\frac{\partial}{\partial x}\psi = p\,\psi$$

$$\hat{H} \doteq i\hbar\frac{\partial}{\partial t}$$

$$\hat{p} \doteq -i\hbar\frac{\partial}{\partial x}$$

Schrödinger Equation

$$E = \frac{p^2}{2m} + U(x) \Rightarrow \quad \hat{H}\,\psi = \left(\frac{\hat{p}^2}{2m} + U(x)\right)\psi$$

$$\boxed{i\hbar \frac{\partial \psi(x,t)}{\partial t} = -\frac{\hbar^2}{2m}\frac{\partial^2 \psi(x,t)}{\partial x^2} + U(x)\psi(x,t)}$$

- Linear partial differential equation with non-constant coefficients; solutions depend on $U(x)$.
- $U(x)$ is a potential energy function describing the interaction of our quantum particle with a much larger system that is essentially fixed.

Definite-Energy States

$$\hat{H}\,\psi_E(x,t) = E\,\psi_E(x,t)$$

$$i\hbar\frac{\partial}{\partial t}\psi_E(x,t) = E\,\psi_E(x,t)$$

$$\psi_E(x,t) = \psi(x)e^{-i\omega t} \qquad \omega = E/\hbar$$

Definite-energy states are also called stationary states because the probability density does not change with time:

$$|\psi_E(x,t)|^2 = |\psi(x)|^2$$

Energy Eigenvalue Equation

- In QM, physical quantities are determined by solving eigenvalue equations; this applies to energy.
- Substitute definite-energy state into SE to arrive at the energy eigenvalue equation, also called the time-independent Schrödinger equation:

$$\boxed{E\,\psi(x) = -\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2} + U(x)\psi(x)}$$

$$\frac{d^2\psi}{dx^2} = -\frac{2m\big(E - U(x)\big)}{\hbar^2}\,\psi$$

Case 1: Constant Potential

- $E > U$, solutions are oscillatory and the wavelength is proportional to $1/\sqrt{E-U}$

$$\frac{d^2\psi}{dx^2} = -\frac{2m(E-U)}{\hbar^2}\psi = -k^2\psi, \quad k \equiv \frac{\sqrt{2m(E-U)}}{\hbar} > 0$$

$$\psi(x) = A\, e^{ikx} + B\, e^{-ikx} = C\sin(kx) + D\cos(kx)$$

$$k = \frac{2\pi}{\lambda} \rightarrow \lambda = \frac{2\pi\hbar}{\sqrt{2m(E-U)}}$$

Case 2: Constant Potential

- $E < U$, solutions are real exponential functions, and the decay length is proportional to $1/\sqrt{U-E}$

$$\frac{d^2\psi}{dx^2} = \frac{2m(U-E)}{\hbar^2}\psi = q^2\psi, \qquad q \equiv \frac{\sqrt{2m(U-E)}}{\hbar} > 0$$

$$\psi(x) = A\, e^{qx} + B\, e^{-qx} = C \sinh(qx) + D \cosh(qx)$$

$$q = \frac{1}{\ell} \;\rightarrow\; \ell = \frac{\hbar}{\sqrt{2m(U-E)}}$$

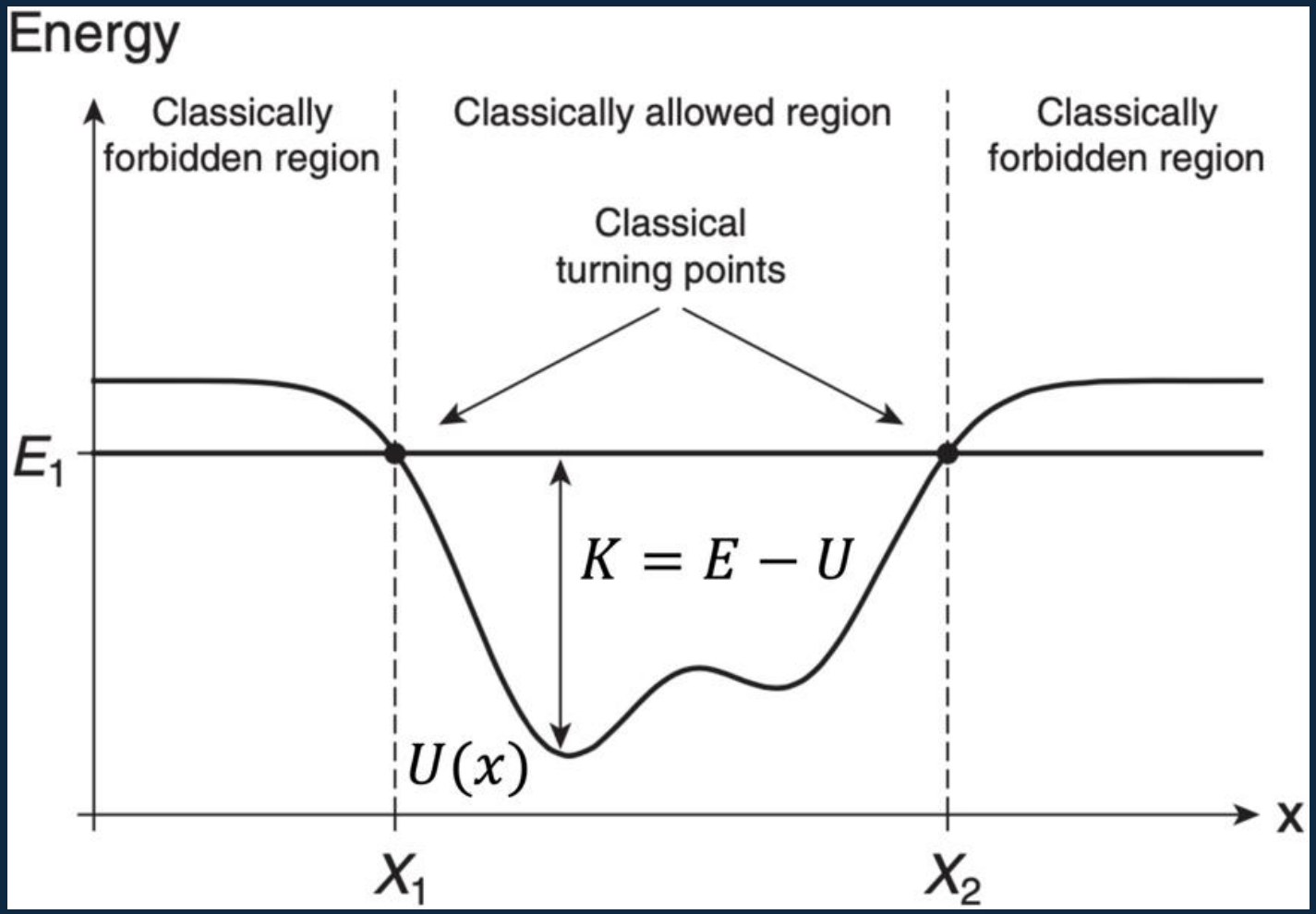
Case 3: $E = U$

- When $E = U$, the solution is linear in x

$$\psi(x) = Ax + B$$

- On a finite interval, this solution can be part of a physical wave function, but if U is constant over all space, this solution is not normalizable.

Energy Diagrams

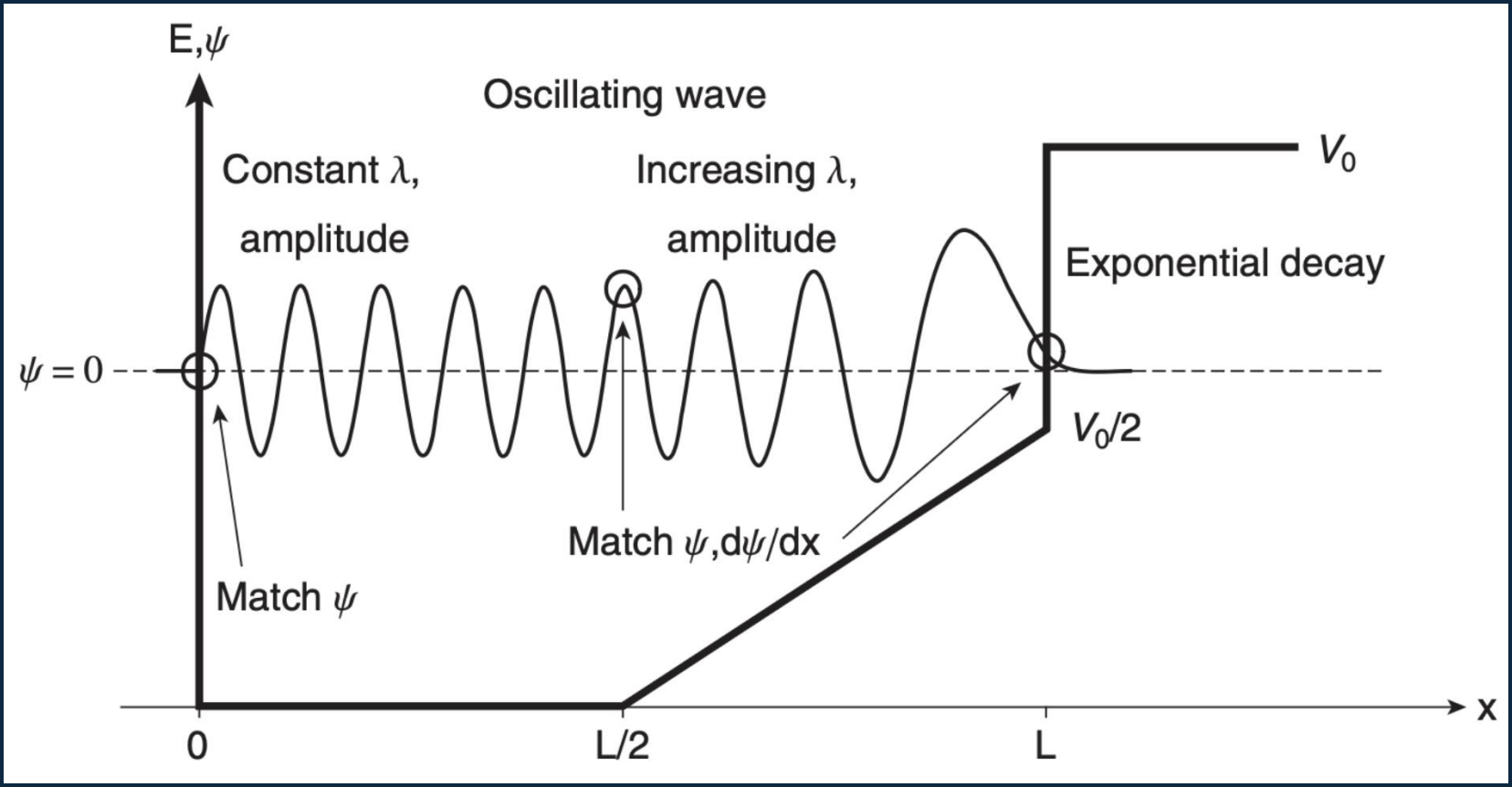


Qualitative Solutions

1(a)	In a classically allowed region, $E > U$, solutions are oscillatory,
1(b)	the local wavelength is proportional to $1/\sqrt{E-U}$,
1(c)	and the local amplitude is large for large wavelengths and small for small wavelengths.
2(a)	In a classically forbidden region, $E < U$, solutions are real exponential functions,
2(b)	and the local decay length is proportional to $1/\sqrt{U-E}$.
3(a)	The time-independent wave function $\psi(x)$ must be continuous everywhere.
3(b)	The first derivative of $\psi(x)$ must be continuous everywhere U is finite.

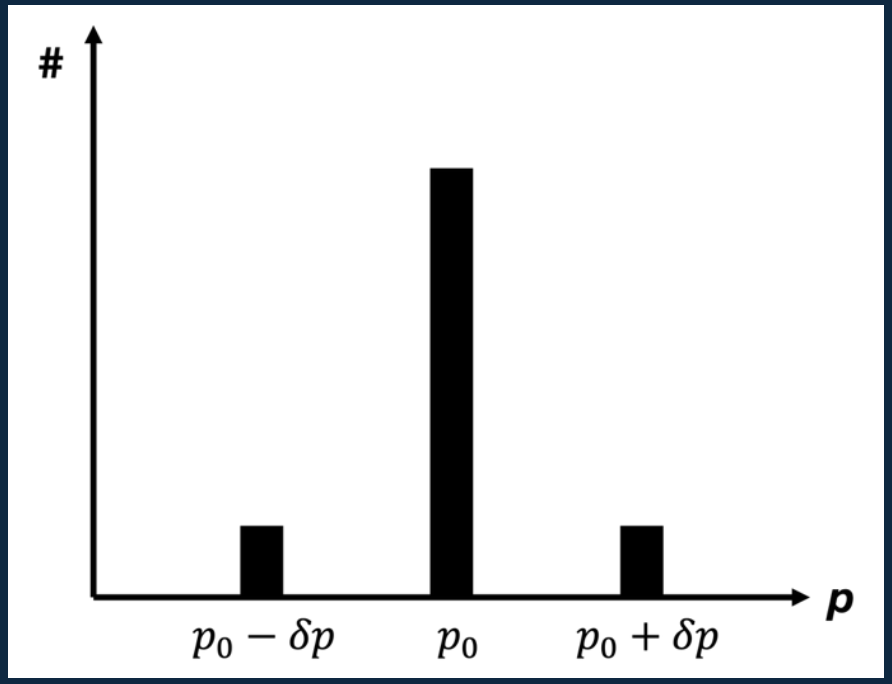
Rules 3(a) and 3(b) are called matching conditions (or boundary conditions), required for second derivative in SE

Qualitative Solutions



Superposition State

- Consider an experiment in which repeated measurements of momentum on identically prepared systems yields the following distribution:



Distribution is sharply peaked: $\delta p \ll p_0$

Superposition State

- The wave function that gives the distribution of measurement outcomes is

$$\psi(x) = A\left[\frac{1}{2}e^{i(k_0-\delta k)x} + e^{ik_0x} + \frac{1}{2}e^{i(k_0+\delta k)x}\right]$$

momentum eigenstates

Superposition State

- For a free particle, momentum eigenstates are also energy eigenstates.
- The possible energies are

$$E_0 = \frac{p_0^2}{2m} = \hbar\omega_0$$

$$E_\pm = \frac{(p_0 \pm \delta p)^2}{2m} = E_0 \pm \frac{p_0}{m}\,\delta p = \hbar(\omega_0 \pm \delta\omega)$$

$$\delta p \ll p_0$$

Superposition State

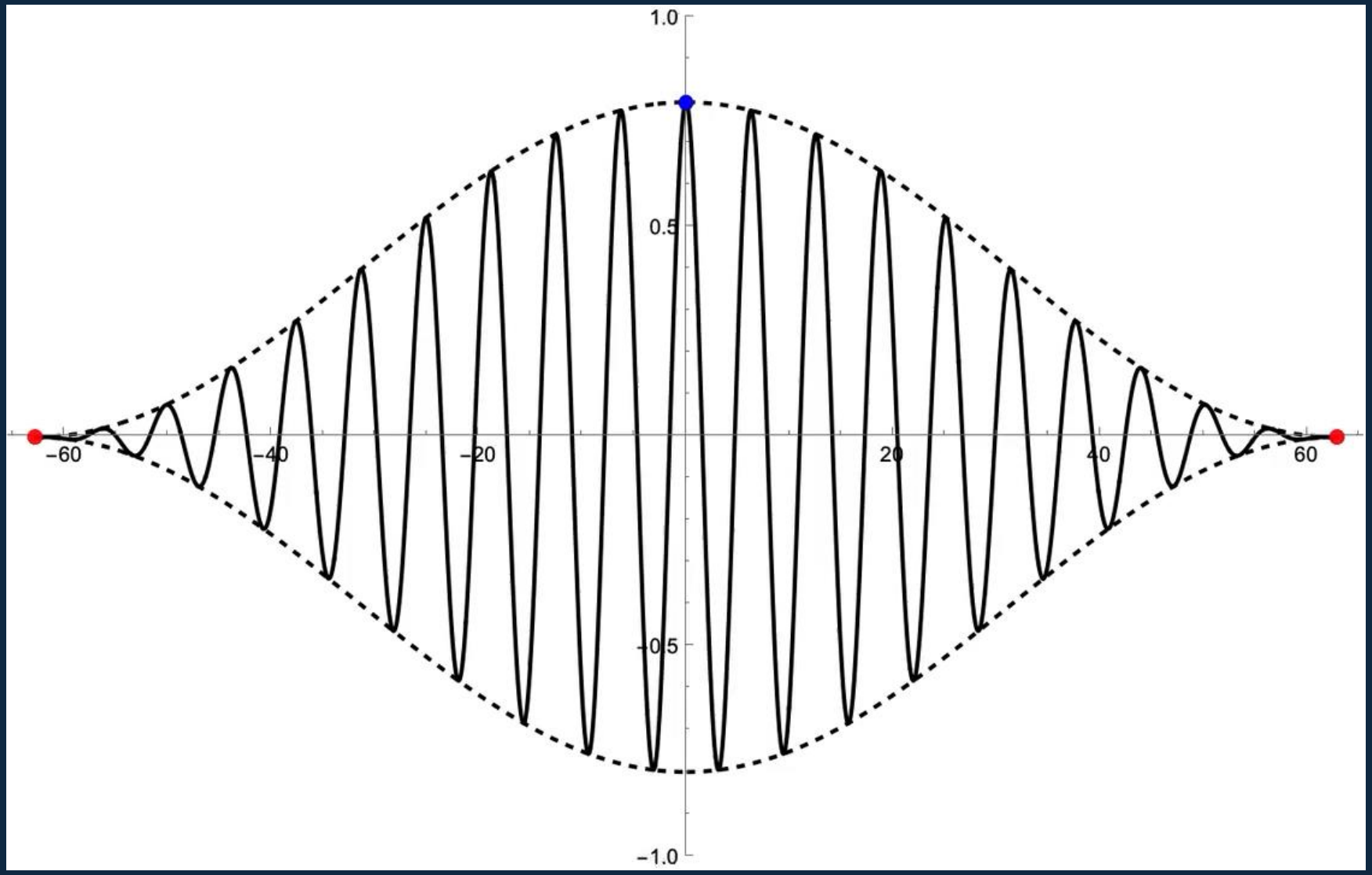
- The full time-dependent wave function is a superposition state:

$$\psi(x,t) = 2A \underbrace{\exp\left[ik_0\left(x - \frac{\omega_0}{k_0}t\right)\right]}_{\text{sinusoidal traveling wave}} \underbrace{\cos^2\left[\frac{\delta k}{2}\left(x - \frac{\delta\omega}{\delta k}t\right)\right]}_{\text{traveling envelope wave}}$$

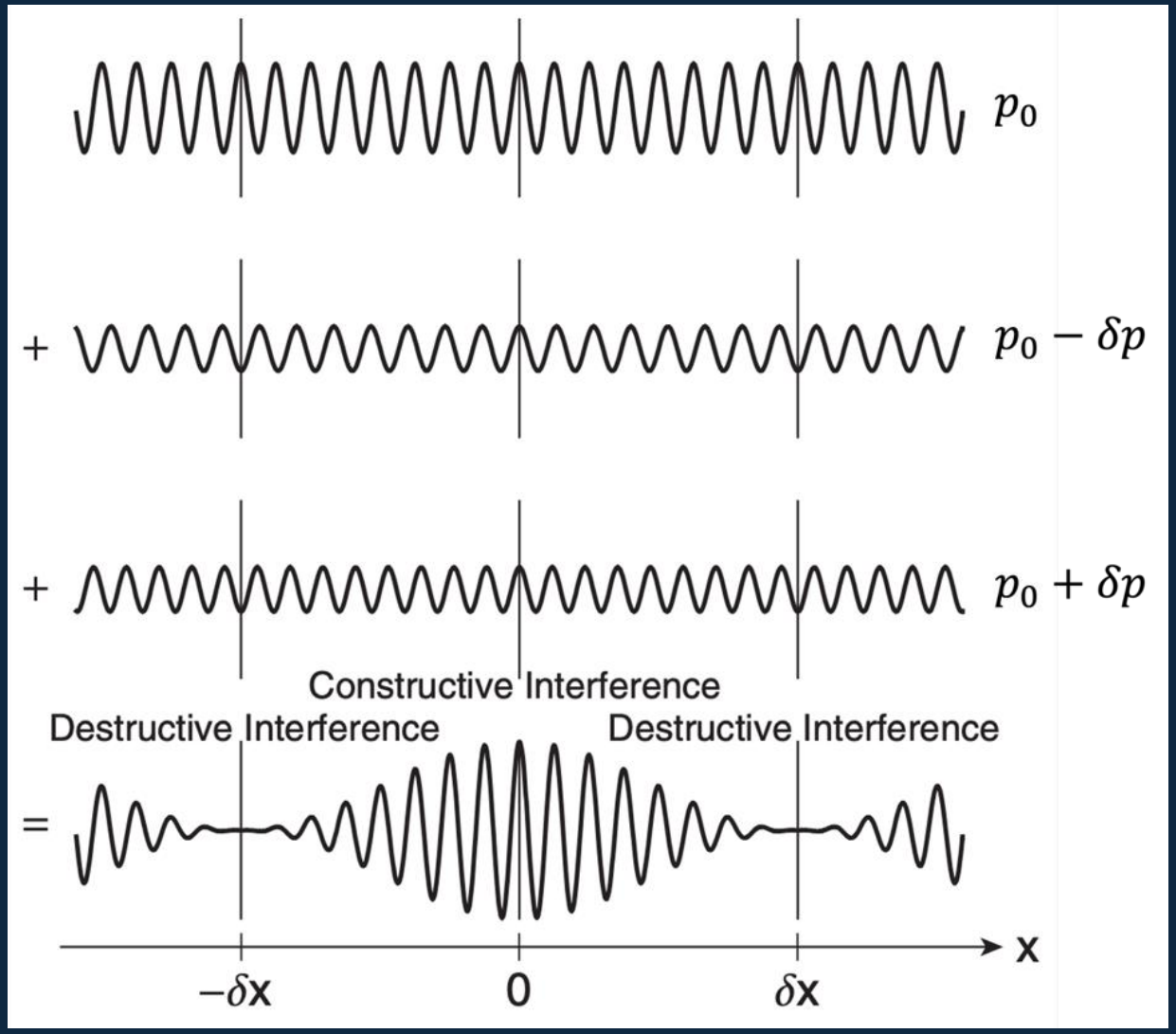
$$\frac{\delta\omega}{\delta k} = \frac{\hbar\delta\omega}{\hbar\delta k} = \frac{p_0 \delta\, p/m}{\delta p} = \frac{p_0}{m}$$

$$\frac{\omega_0}{k_0} = \frac{\hbar\omega_0}{\hbar k_0} = \frac{p_0^2/2\,m}{p_0} = \frac{p_0}{2m}$$

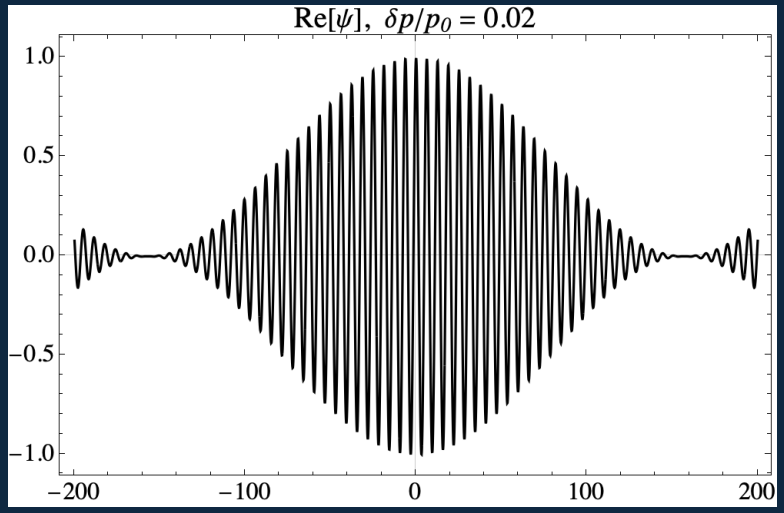
Wave Packet

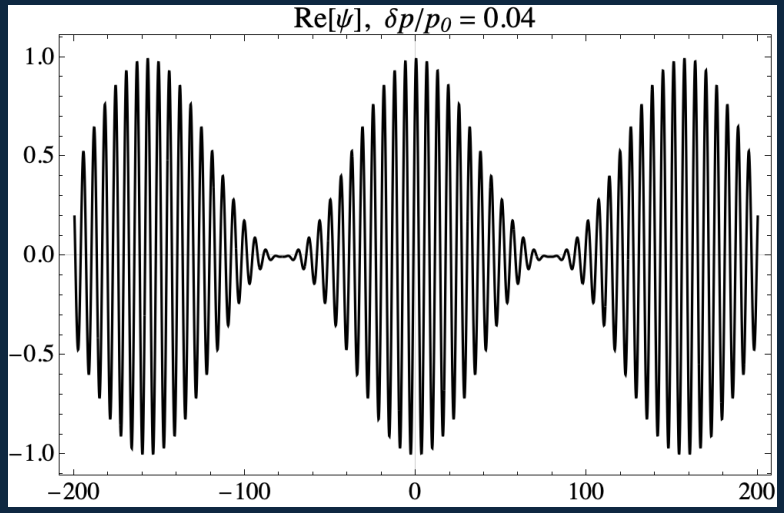


Superposition State



Uncertainty Relation





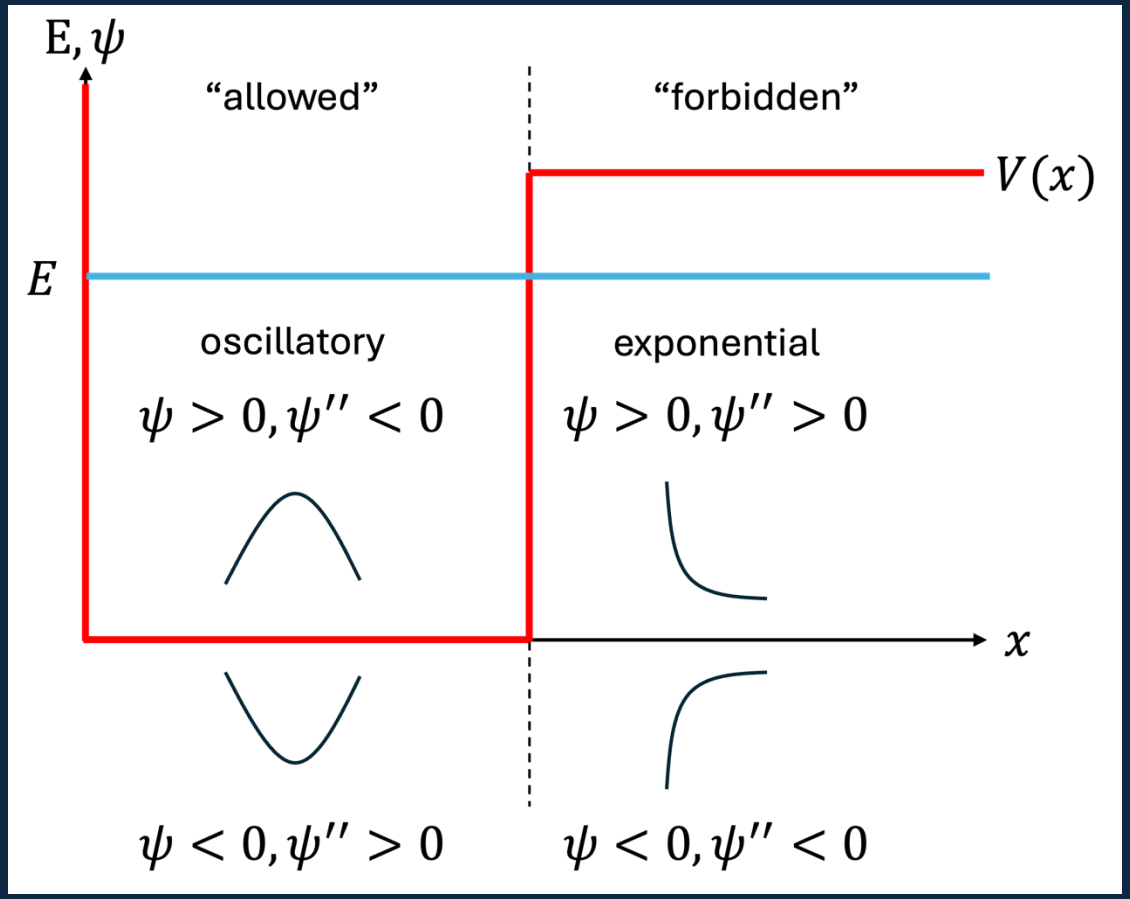
$$\frac{\delta p}{2\hbar}\delta x = \frac{\pi}{2} \Rightarrow \delta x\, \delta p = \pi\hbar$$

$$\Delta x\, \Delta p \geq \frac{\hbar}{2}$$

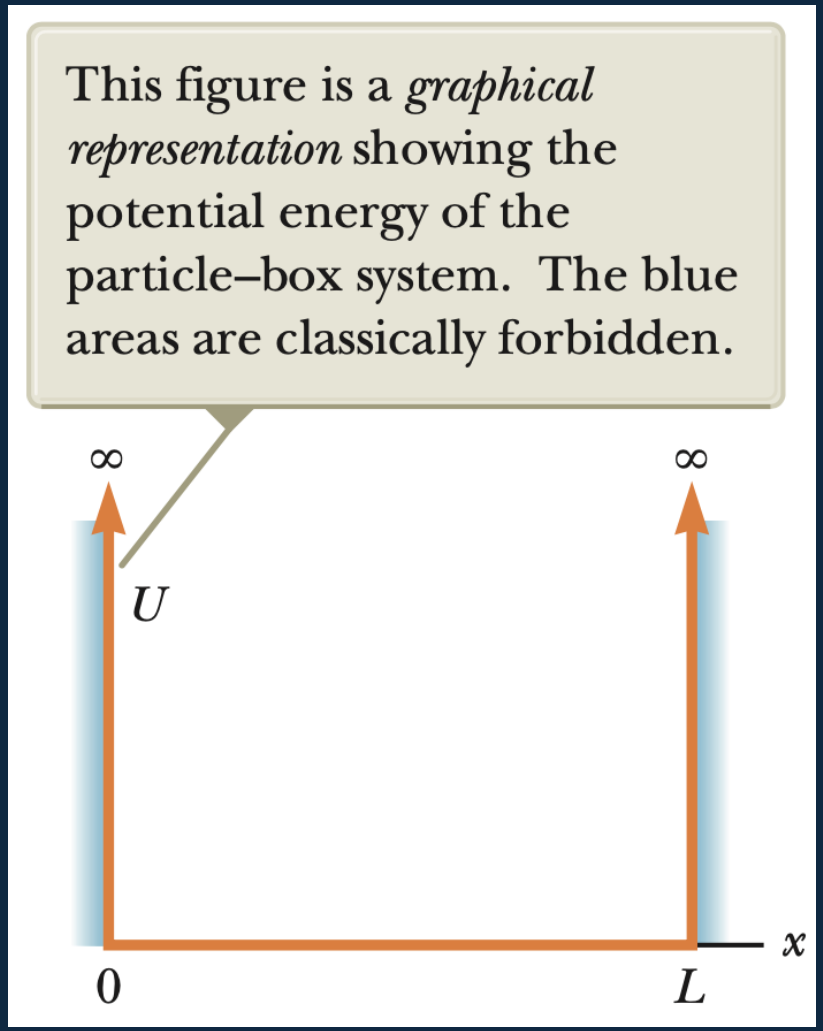
The uncertainty relation is a statement about distributions; it does not mean that individual measurements are necessarily imprecise.

Bound States in Potential Wells

$$\psi'' = -\frac{2m\big(E - U(x)\big)}{\hbar^2}\,\psi$$

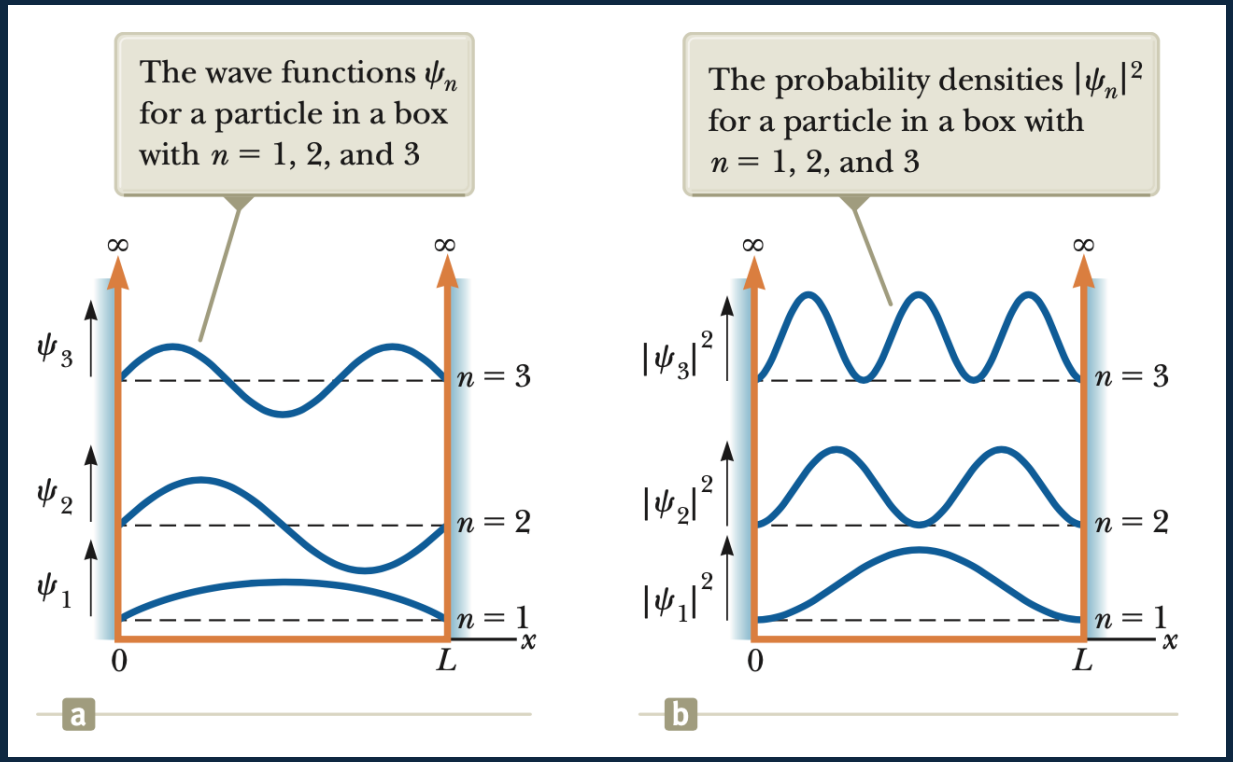


Infinite Square Well



Infinite Square Well

$$\psi_n(x) = \sqrt{\frac{2}{L}} \sin\frac{n\pi x}{L}, \qquad E_n = \frac{n^2\pi^2\hbar^2}{2mL^2}, \qquad n = 1, 2, 3, \ldots$$



Example: Infinite Square Well

A particle in a box is in its ground state. Calculate the probability that the particle will be found in the region

(a) $0 < x < L/2$,

(b) $0 < x < L/3$,

(c) $0 < x < 3L/4$.

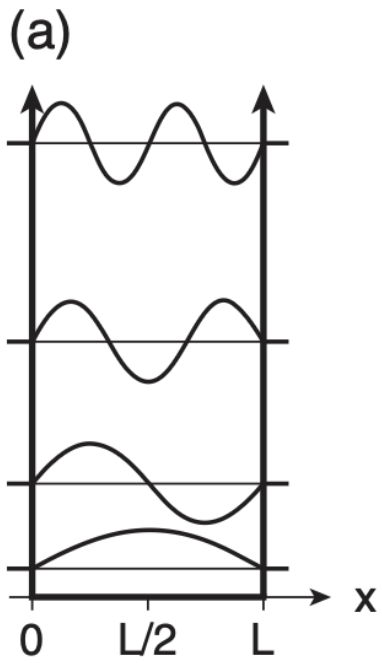
(d) Estimate the probability of finding the particle in a *very small region* $\Delta x = 0.01L$ centered on position $x = 3L/8$.

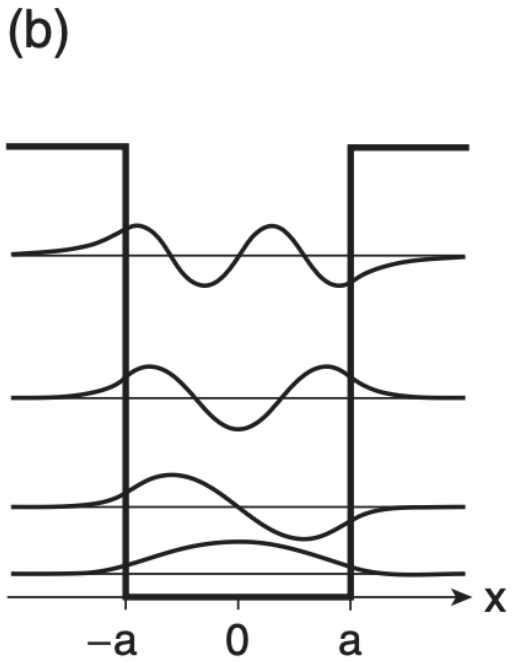
Properties of Bound Energy Eigenstates in Potential Energy Wells

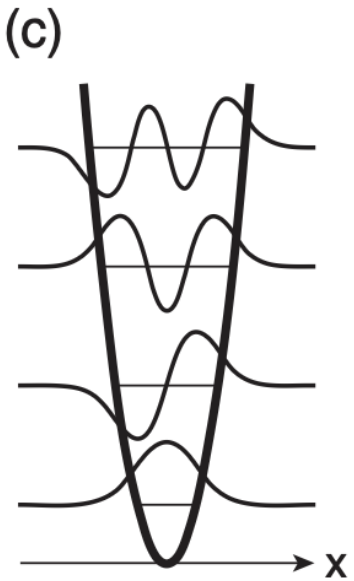
- The nth energy eigenstate has n antinodes and $n - 1$ nodes.
- For a symmetric potential well, eigenfunctions are either symmetric (even) or antisymmetric (odd) with respect to the center of the well.
- The symmetric states are said to have “even parity” and the antisymmetric states have “odd parity.”
  - The parity of the energy eigenstates alternates with successive energy levels.

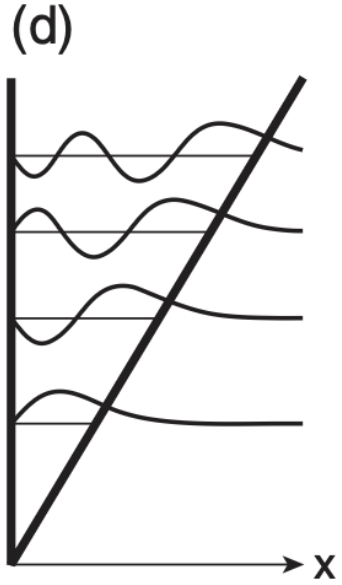
Examples

(a) Infinite square well
(b) Finite square well
(c) Harmonic oscillator
(d) Linear potential

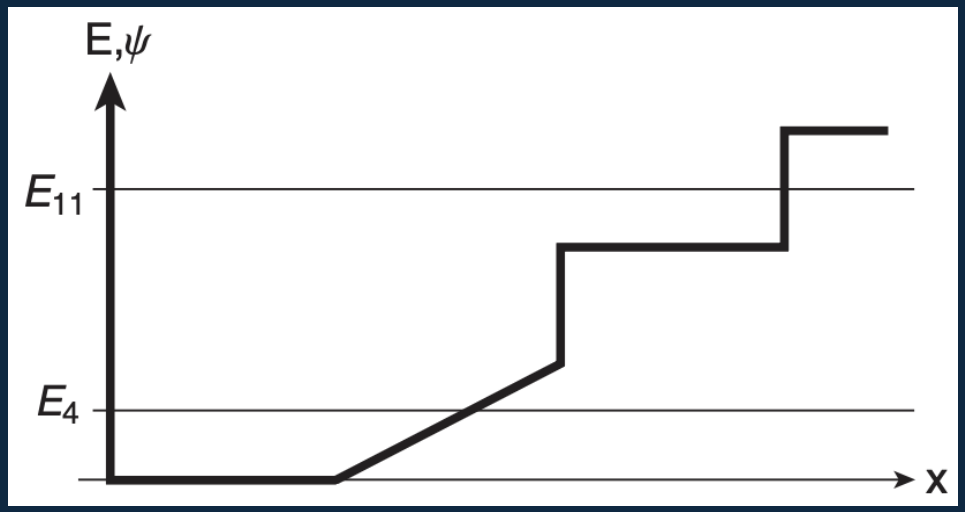


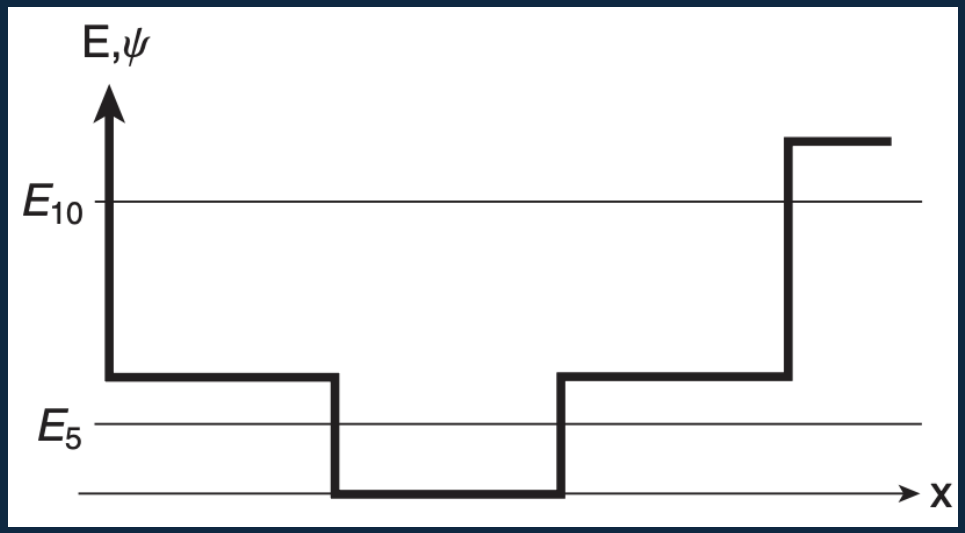




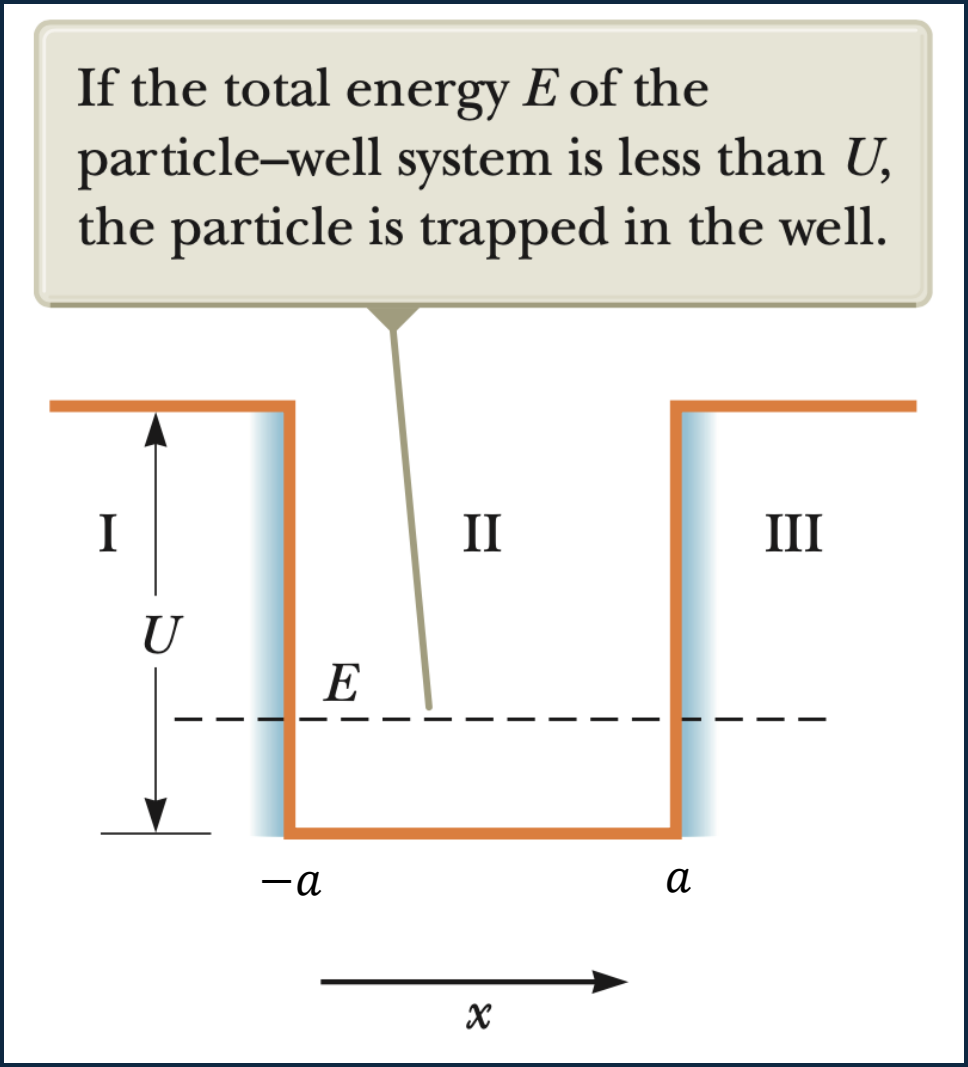


Sketching Bound States





Finite Square Well



Finite Square Well

- General strategy: write the solution for each region in terms of integration constants and determine the full solution using boundary conditions and normalization.

$$\psi(x) = \begin{cases} A\, e^{qx} + B\, e^{-qx}, & x < -a \\ C \sin(kx) + D \cos(kx), & -a < x < a, \\ F\, e^{qx} + G\, e^{-qx}, & x > a \end{cases}$$

- Including energy E, there are seven "unknowns"
- 2 boundary conditions at $x = \pm\infty$
- +4 matching conditions (2 at $x = -a$ and 2 at $x = a$)
- +1 normalization condition
- = 7 independent constraints

Finite Square Well

- Let's focus on the even solutions:

$$\psi_{\text{even}}(x) = \begin{cases} A\, e^{qx}, & x < -a \\ D\cos(kx), & -a < x < a, \\ Ae^{-qx}, & x > a \end{cases}$$

$$\psi_{\text{even}}(a): \quad D\cos(ka) = Ae^{-qa}$$
$$\psi'_{\text{even}}(a): \quad -kD\sin(ka) = -qAe^{-qa}$$

- Quantization conditions:

$$k\tan(ka) = q \qquad \text{even solutions}$$

$$-k\cot(ka) = q \qquad \text{odd solutions}$$

Example: Finite Square Well

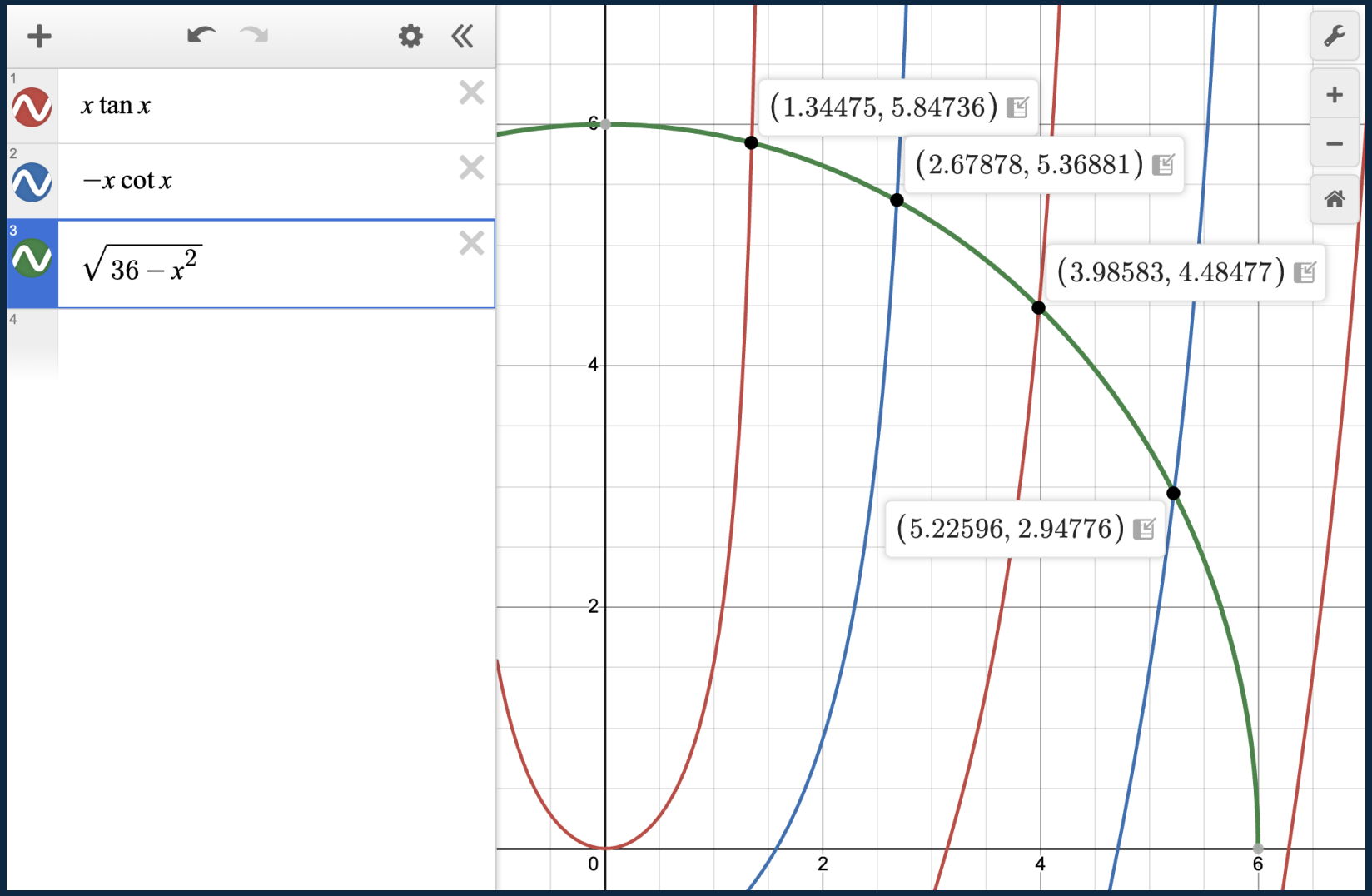
A finite square well has a width of $2a$ and a depth of

$$U_0 = 36\frac{\hbar^2}{2ma^2}$$

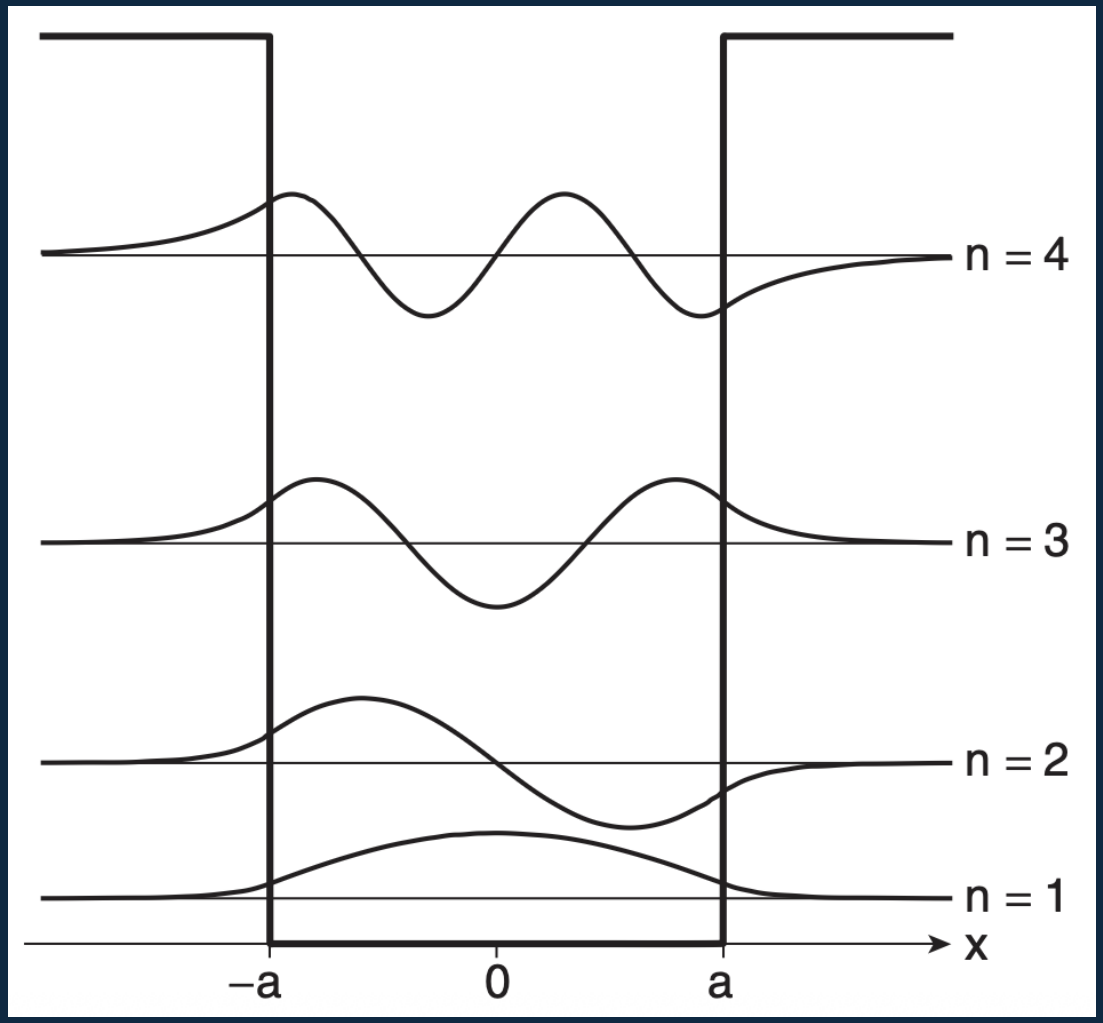
(a) Find the number of bound energy eigenstates and their corresponding energies.
(b) Time permitting, determine the number of bound states in a finite square well of any depth.

Hint: use *natural units* where $\hbar = 2m = a = 1$ and find the corresponding units for energy and momentum. This will make it easier to solve the transcendental equations using a computer.

Graphical Solution



Finite Square Well



Conceptual Question

For which of the following well depths does the ground state have the lowest energy?

A. U_0

B. $2U_0$

C. $3U_0$

D. $4U_0$

Conceptual Question

Which of the following statements are true regarding the energy levels in a finite well of fixed depth? Select all that apply.

A. The larger $U_0 - E$, the lower the energy compared with the infinite well

B. The higher the excited state, the lower the energy compared with the infinite well

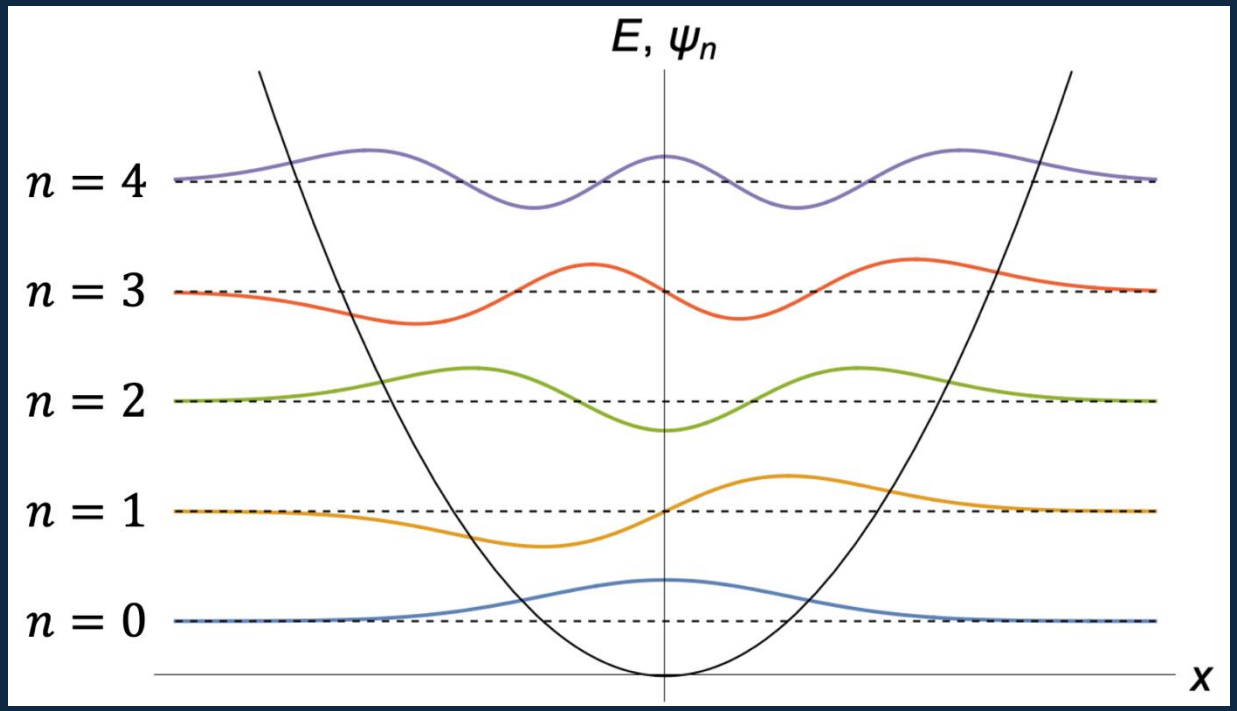
C. An infinite number of bound state energy levels exist

Conceptual Question

Which of the following statements are true regarding the energy eigenfunctions in a finite well of fixed depth? Select all that apply.

A. The higher the energy, the less likely the particle will be found outside the well
B. The lower the energy, the more likely the particle will be found inside the well
C. Only a finite number of bound state energy eigenfunctions exist

Quantum Harmonic Oscillator

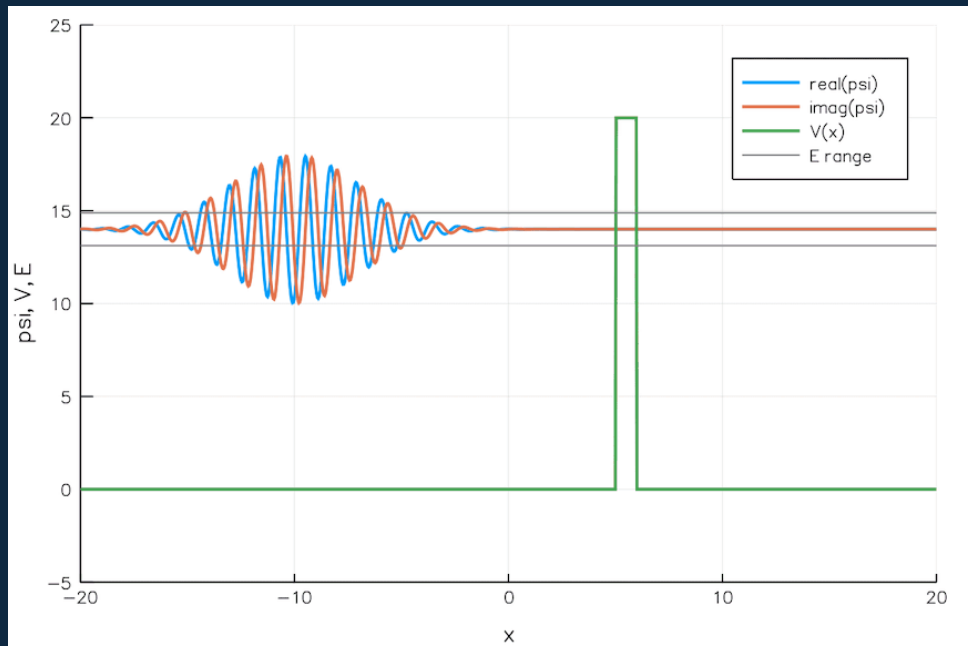


$$E_n = \hbar\omega\left(n + \frac{1}{2}\right), \qquad n = 0, 1, 2, \ldots$$

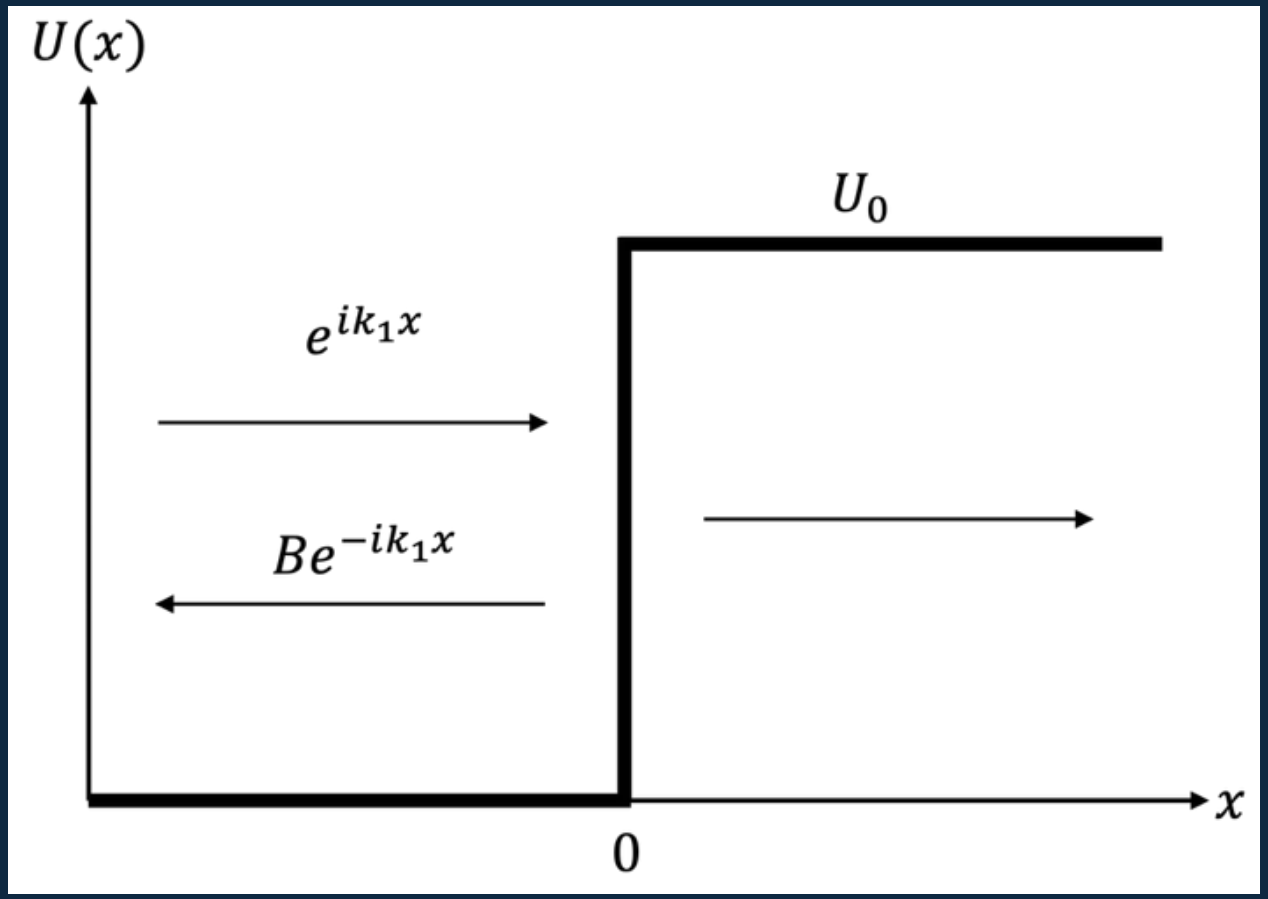
Scattering Theory

- Distant free particle travels to a region with a localized change in the potential energy function. Interaction produces nonzero amplitudes for the wave packet to both transmit and reflect.
- Consider wave packet much larger than width of region where the potential changes significantly; momentum eigenstates work to good approximation (also easy to analyze).
- Think of particles being continually projected onto a target; some reflect, and some transmit.

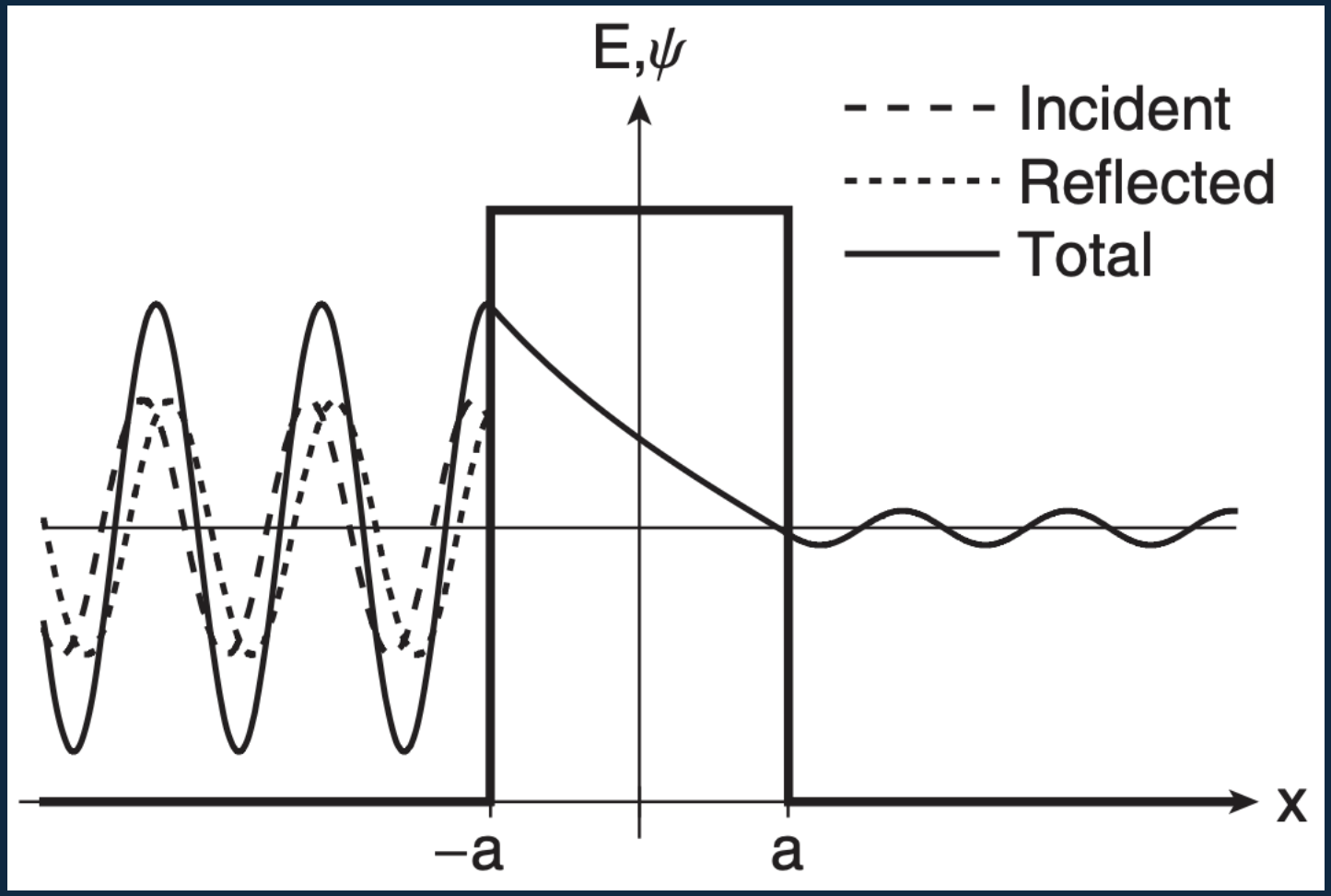
A simulation of a wave packet incident on a potential barrier.



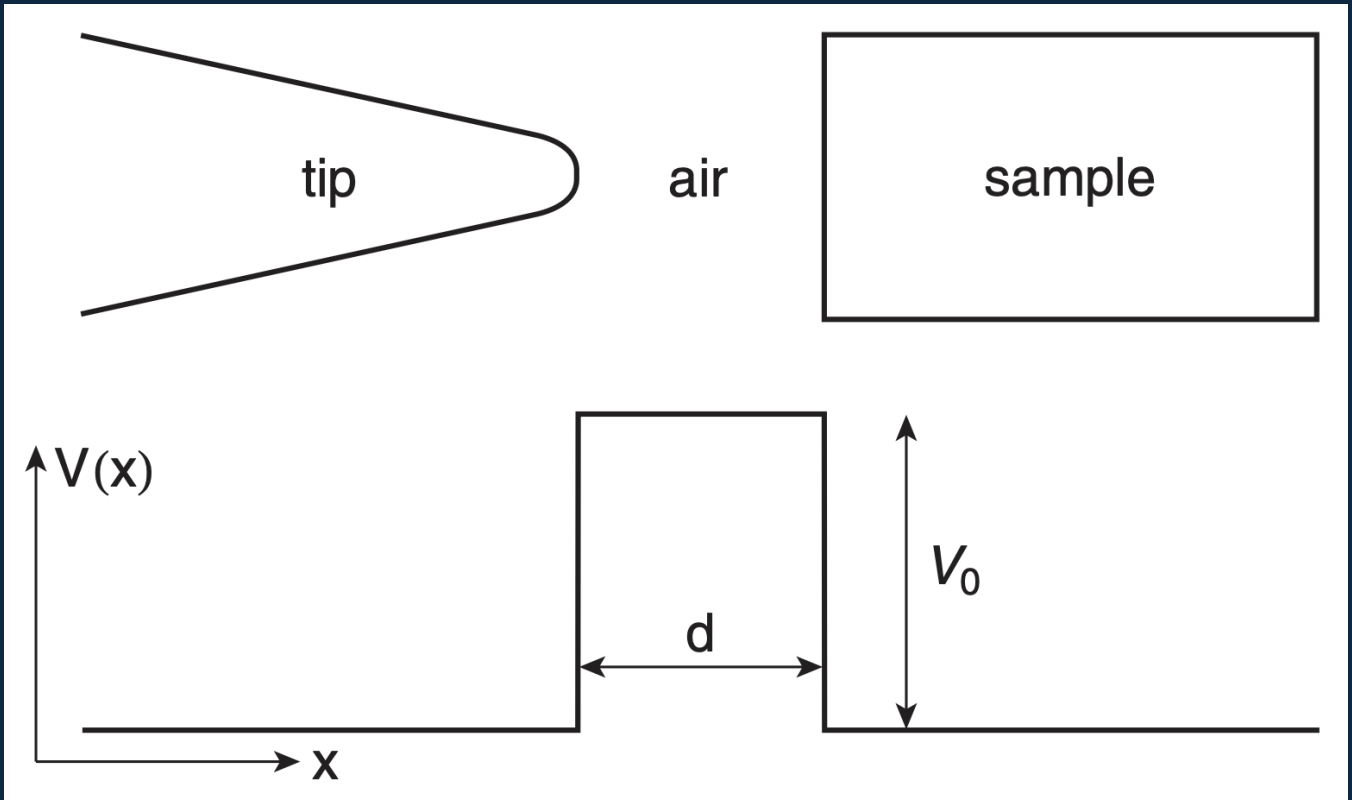
Step Potential



Tunneling Through a Barrier

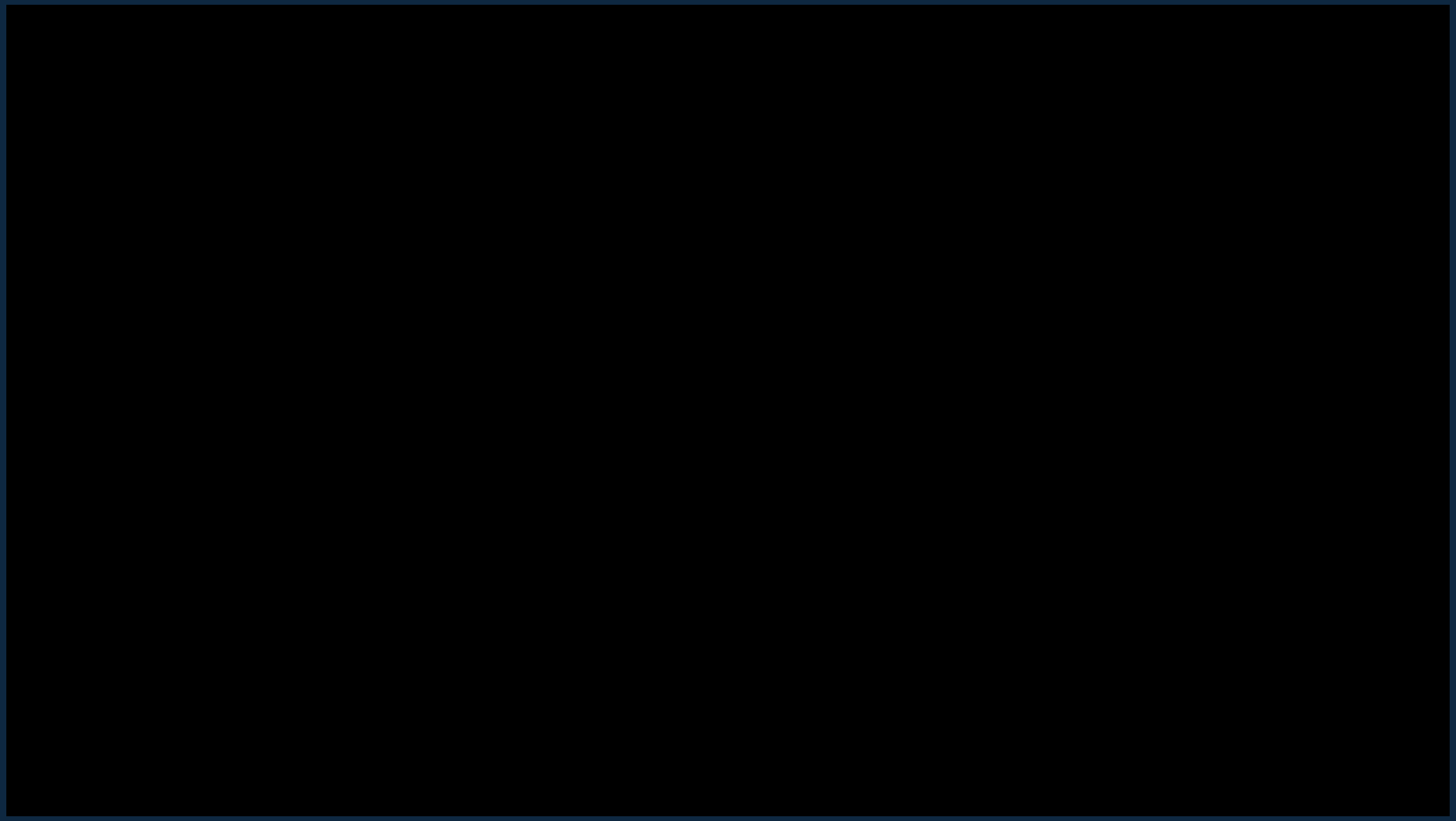


Scanning Tunneling Microscope



$$I = I_0 \, e^{-2qd}$$

A Boy And His Atom



THE NOBEL PRIZE IN PHYSICS 2025



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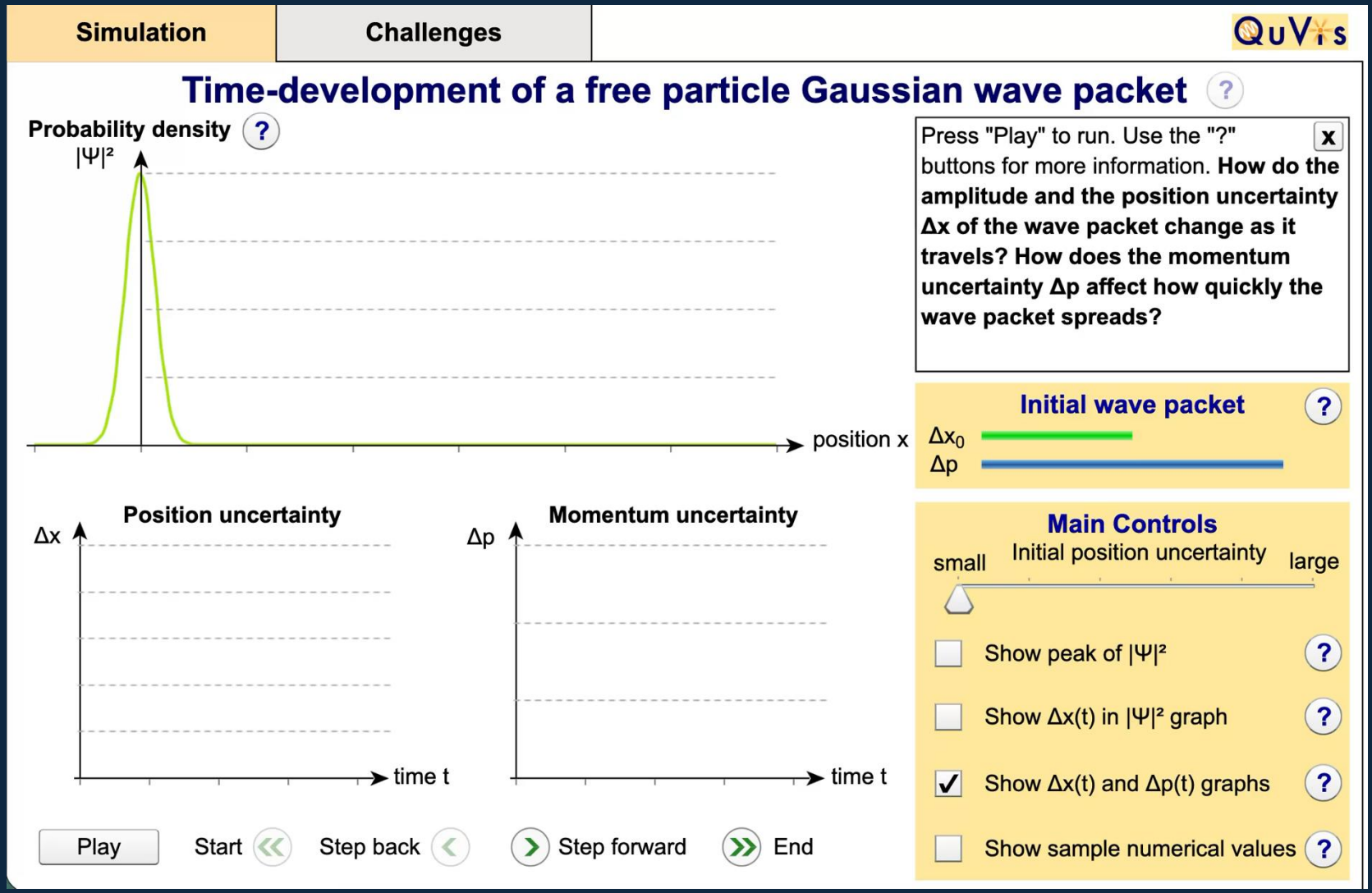
"for the discovery of macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit"

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The Classical Limit

- The classical limit of quantum mechanics refers to predictions of quantum mechanics for systems with dynamical magnitudes that are large compared to $\hbar$.
- *Classical* and *macroscopic* are distinct terms; e.g., superconductivity is a macroscopic quantum phenomenon. By that same token, *quantum* and *microscopic* are also distinct terms.
- Correspondence principle: the probability distributions of quantum mechanics should approach the probability distributions of an *ensemble of classical trajectories* in the classical limit.

Free Particle Evolution



Ehrenfest's Theorem

- From the Schrödinger equation, for any wave function,

$$\frac{d\langle\hat{x}\rangle}{dt} = \frac{\langle\hat{p}\rangle}{m} \qquad \frac{d\langle\hat{p}\rangle}{dt} = \left\langle -\frac{dU}{dx} \right\rangle = \langle F(x)\rangle$$

- If the width of the position probability distribution is small compared to distances over which the force varies, or the force is a linear function of position, then

$$\langle F(\hat{x})\rangle \approx F(\langle\hat{x}\rangle)$$

and expectation values follow classical mechanics.

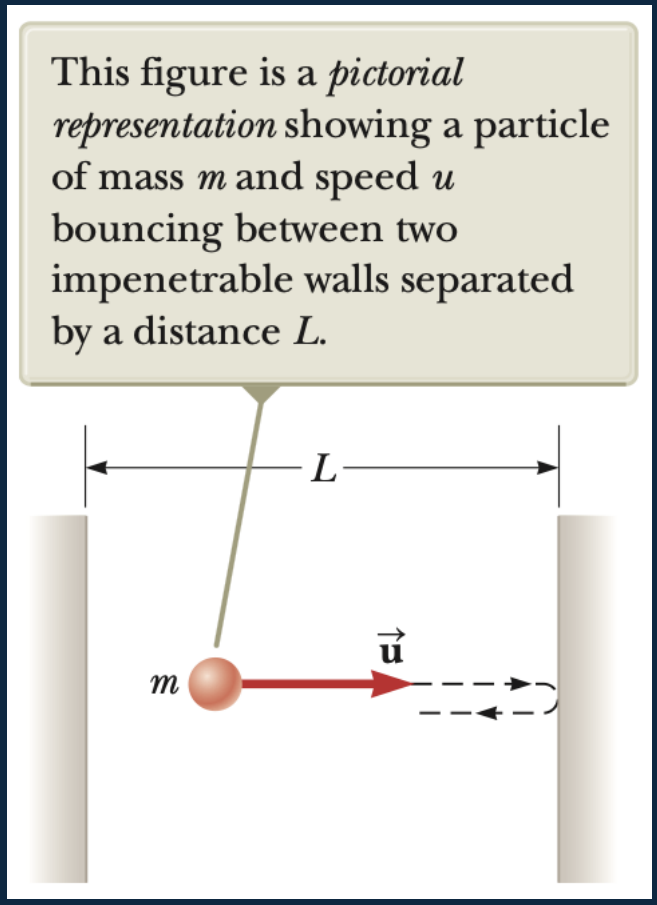
Neither necessary nor sufficient for classical behavior.

The Classical Limit

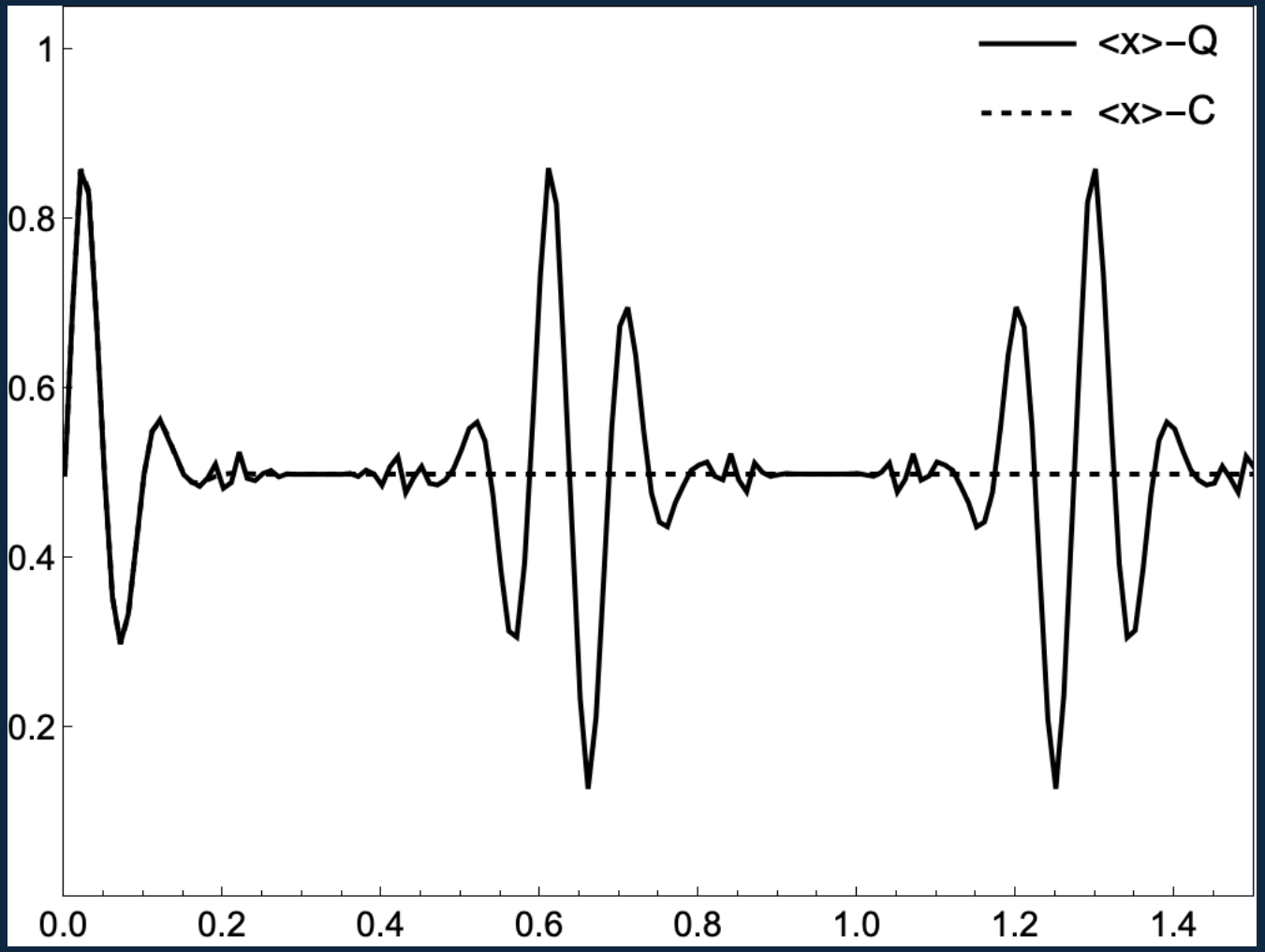
- Consider an infinite square well
- Classical physics: let's say we know precisely the speed of the ball, but we lost track of its position. We can estimate the probability to find it within a small interval dx as

$$\rho\, dx = \frac{dt}{T} = \frac{dx/u}{L/u} = \frac{1}{L}\, dx \Rightarrow \quad \rho = \frac{1}{L}$$

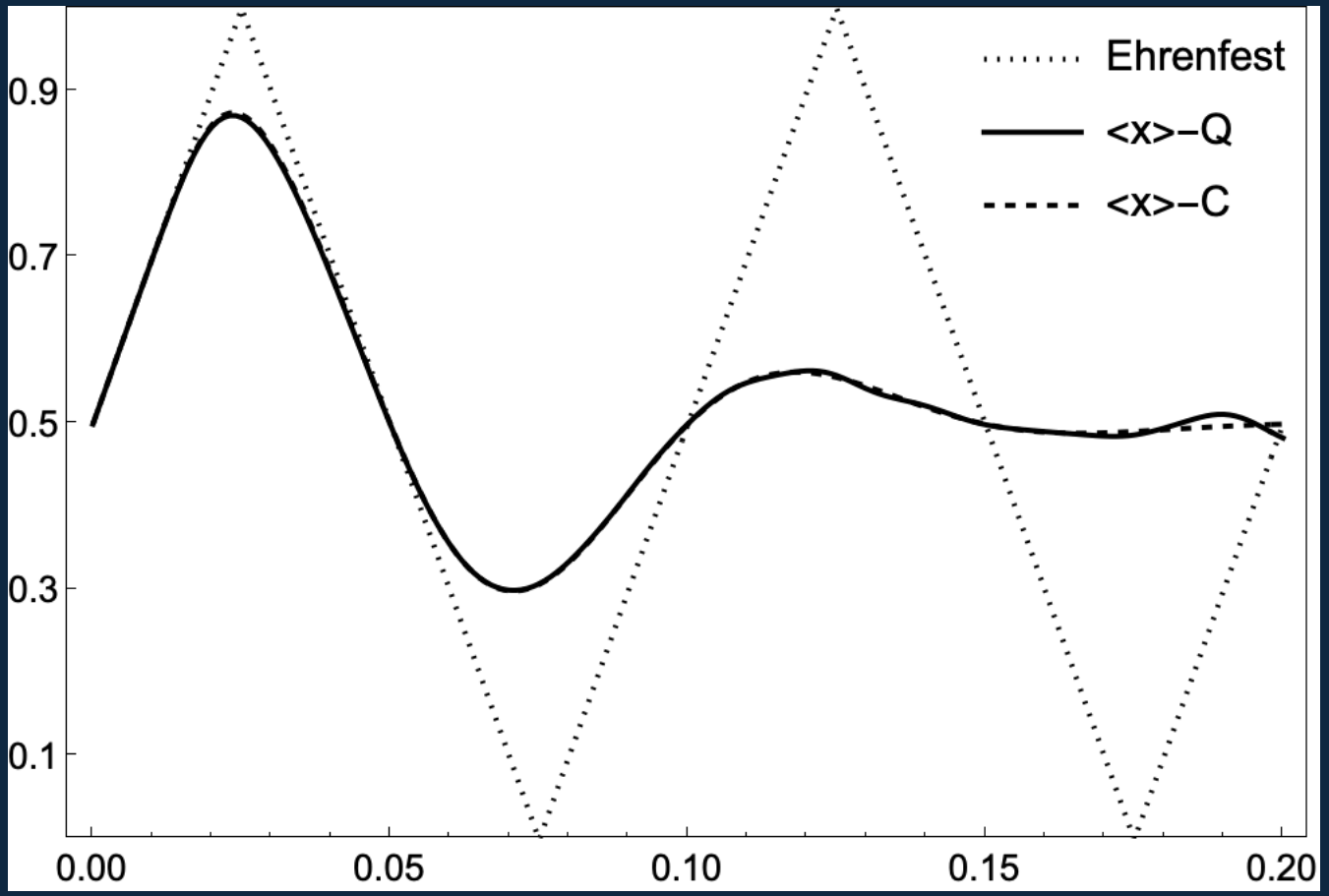
- Uniform distribution; equally likely to find it anywhere inside the well
- If there was any uncertainty about the velocity of the ball, the exact calculation is more complicated—statistical mechanics!



The Classical Limit



The Classical Limit



Harmonic Oscillator

- Steady-state classical probability distribution:

$$T = \frac{\pi}{\omega}$$ Half-period, time spent between turning points

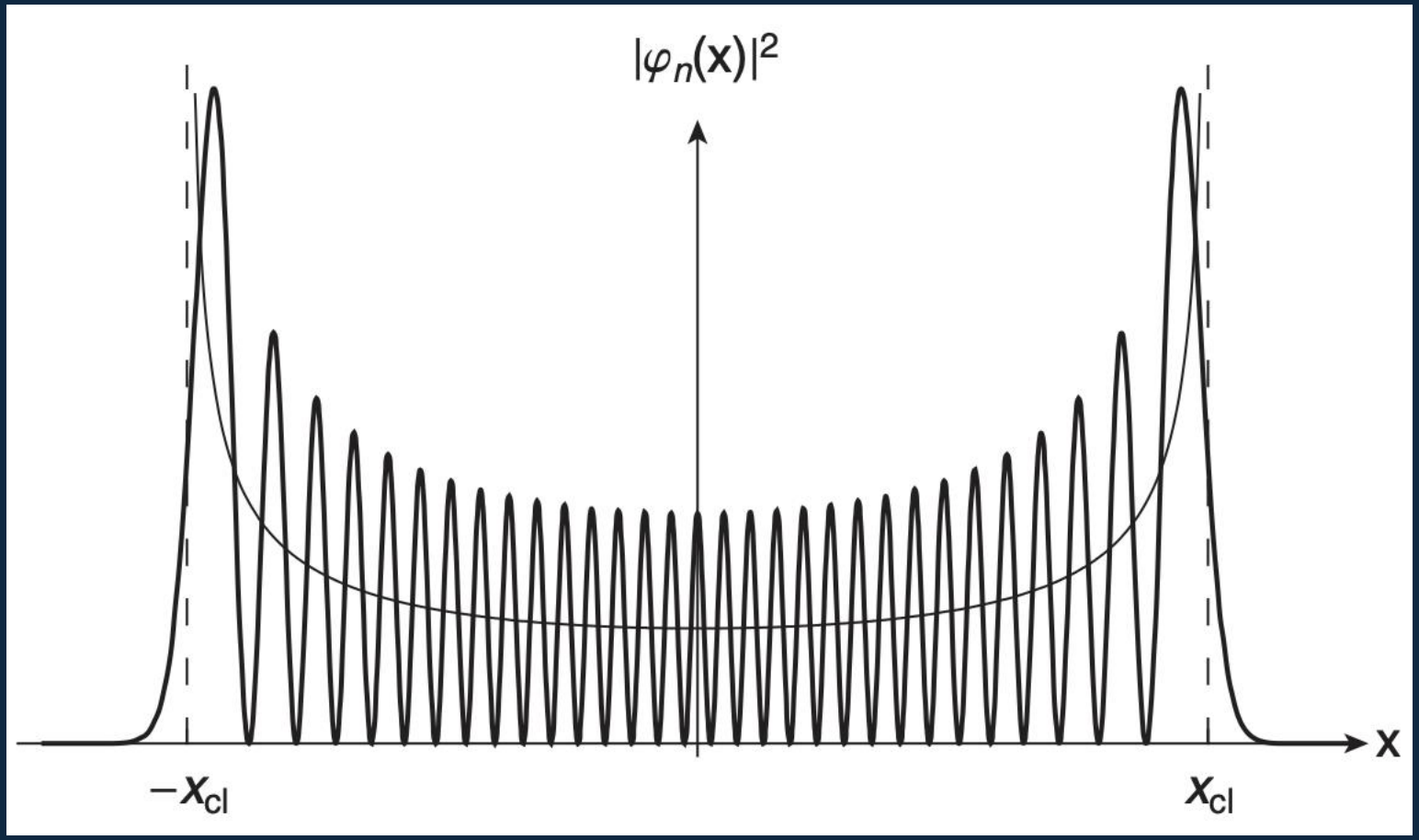
$$\rho\, dx = \frac{dt}{T} \Rightarrow \rho = \frac{1}{T}\frac{dt}{dx} = \frac{1}{v(x)\, T}$$

$$E = \frac{1}{2} m\, v^2 + \frac{1}{2}\, m\omega^2\, x^2 \Rightarrow v(x) = \omega\sqrt{A^2 - x^2}$$

$$\rho = \frac{1}{\pi\sqrt{A^2 - x^2}}$$

$$A = \sqrt{\frac{2E}{m\omega^2}}$$

The Classical Limit



Energy eigenfunction of the harmonic oscillator ($n = 30$): the coarse-grained quantum distribution matches the classical distribution; however, it does not converge point wise!

Entanglement

- Informally, entanglement happens when the state of a system cannot be neatly factored into independent parts
- Superposition states can exist for composite systems, and they encode joint probability distributions
- Measurement outcomes for one particle can be correlated with measurement outcomes for another if the two particles are entangled
- Two systems become entangled by interacting with each other. The “entangling interactions” are always local.



Multi-electron atoms

- To a good approximation, we can use $\psi(\mathbf{r})$ to describe a single electron in the Coulomb potential of hydrogen
- Schrödinger realized the wave function for helium depends on positions of both electrons.
- $\psi(\mathbf{r}_1, \mathbf{r}_2)$ is a wave function over configuration space.
- In famous letter to Max Born (1926), Einstein writes "God does not play dice" and expressed concerns with Schrödinger's waves in "3n-dimensional space"
- The misinterpretation of ψ as a physical wave in ordinary space is only possible for one-particle states!
- $|\psi(\mathbf{r}_1, \mathbf{r}_2)|^2$ is the probability density in configuration space for particle 1 being at $\mathbf{r}_1$ and particle 2 being at $\mathbf{r}_2$.

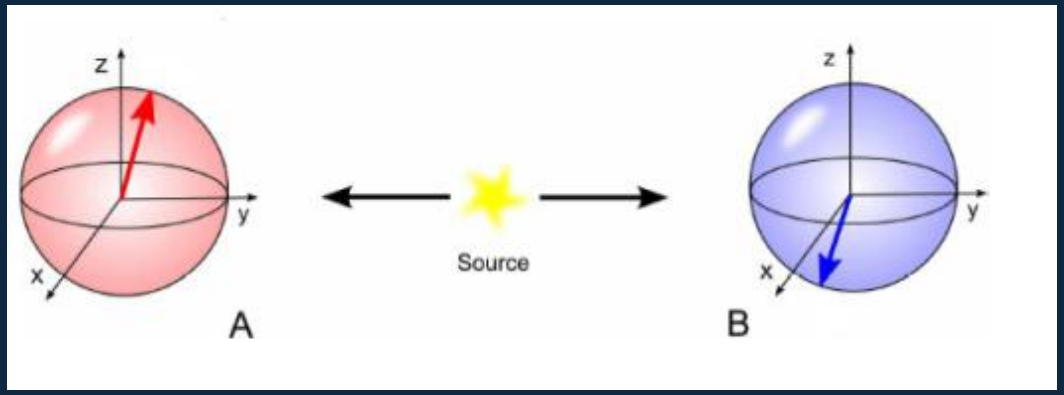
depends on $\mathbf{r}_2$ depends on $\mathbf{r}_1$ correlations!

EPR Paper (1935)

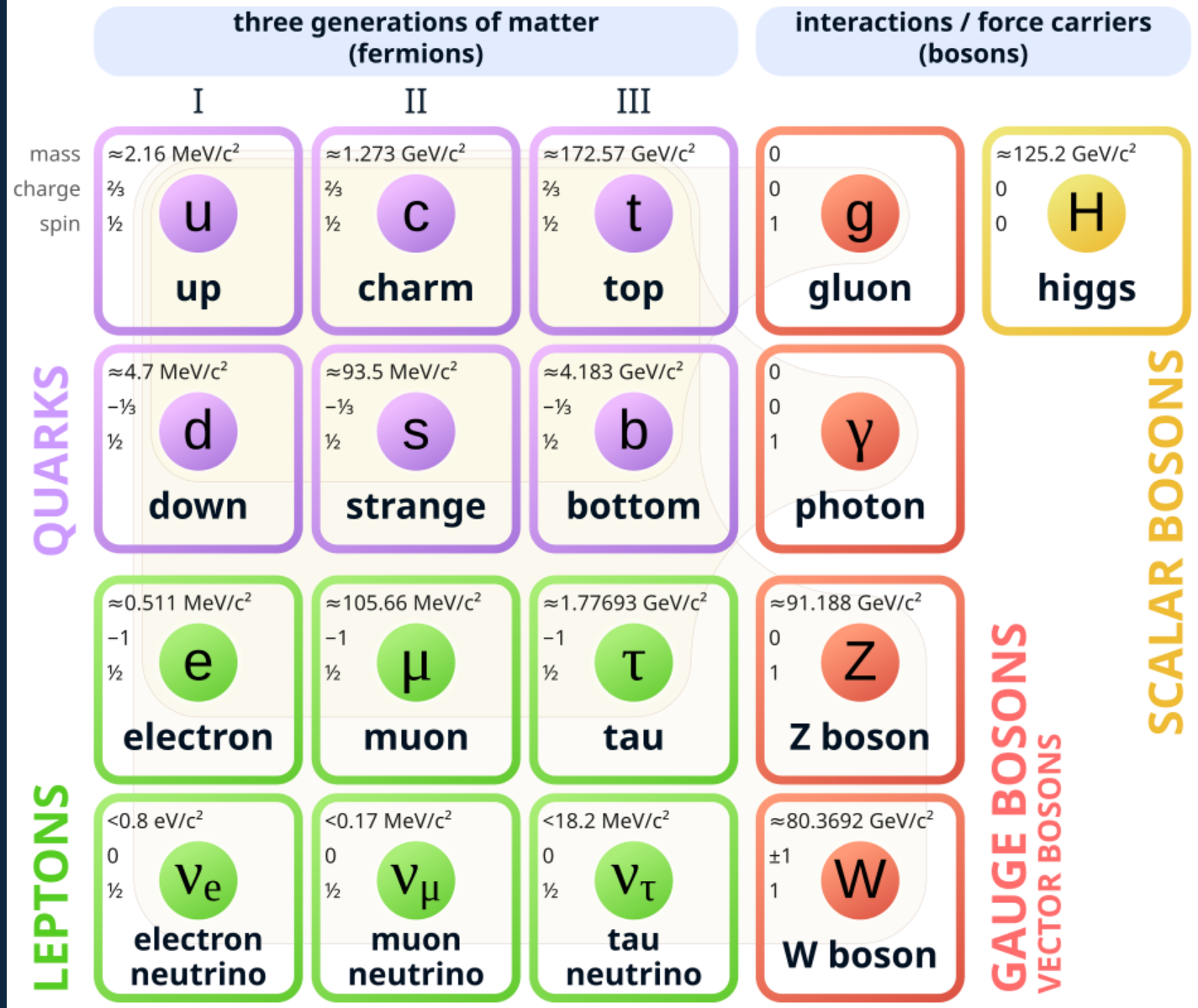
- Einstein-Podolsky-Rosen (EPR) paper argues that the description of physical reality provided by quantum mechanics is incomplete.
- Demonstrates that “elements of reality” exist that are not part of quantum theory and speculates that it should be possible to construct a theory containing “hidden variables.”
- Using an entangled two-particle state, they argue that if no faster-than-light influence is possible, then each particle must have definite properties before measurement.**
- Since quantum mechanics does not generally assign definite properties prior to measurement, it must be incomplete.
- While this is the crux of their argument, they go further, arguing that even *incompatible observables* must have definite pre-existing values!

EPR-Bohm Thought Experiment

- Fundamental particles have a property called intrinsic spin, which is a form of angular momentum that is not associated with their motion through space.
- In the EPR-Bohm thought experiment, a particle with zero spin (like a neutral pion) decays into two particles, A and B, each with spin-1/2 (electron + positron).
- Spin angular momentum must be conserved during the decay, so particles A and B have opposite spin.



Standard Model of Elementary Particles

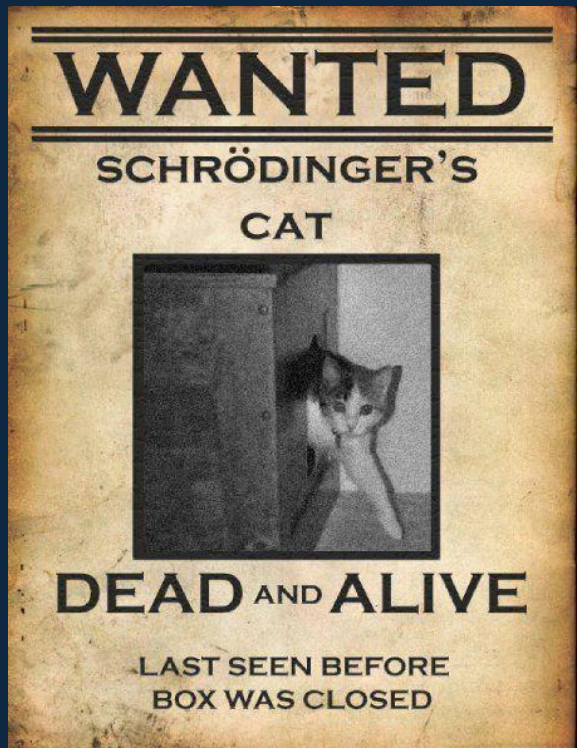


Schrödinger's Cat

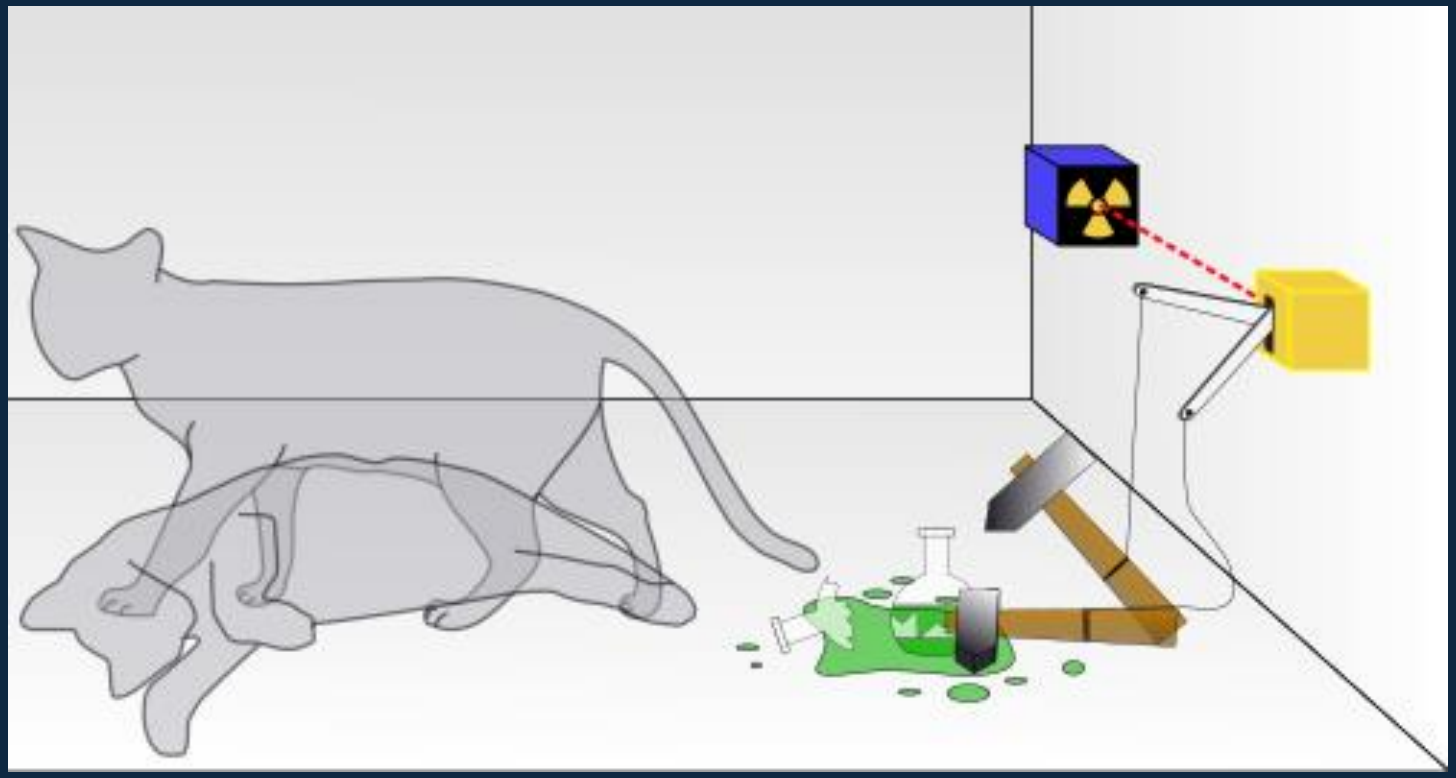
Shortly after the EPR paper was published, Schrödinger introduced the term entanglement and emphasized that "non-separability" was a general feature of quantum mechanics.

He and Einstein agreed on the strangeness of a theory in which systems lack independent states after interaction, and physical properties are not straightforwardly defined prior to measurement.

Introduced the famous "Schrödinger's cat" thought experiment to show that the standard interpretation leads to strange consequences if taken literally at all scales.



Schrödinger's Cat



Bell's Theorem

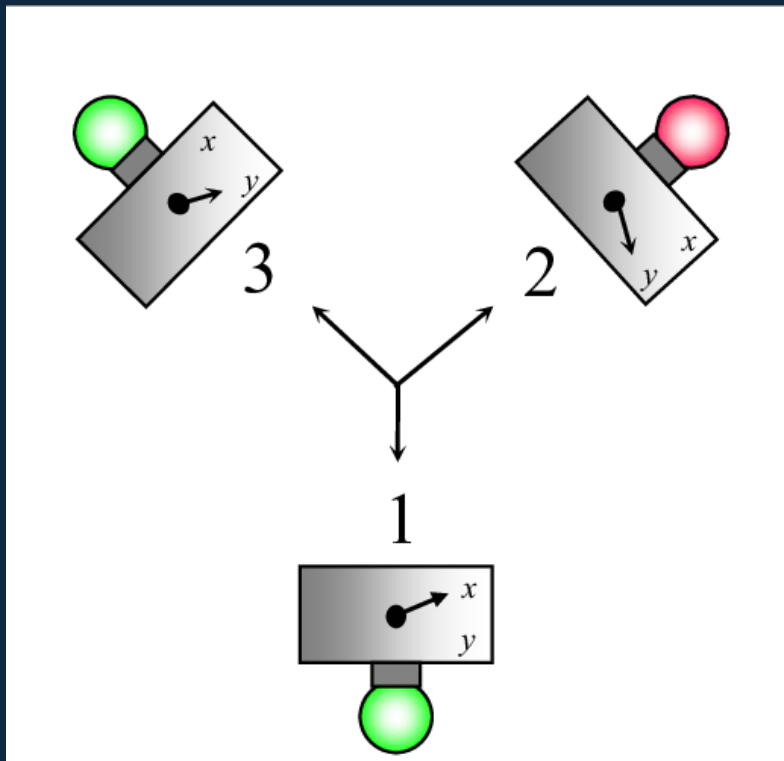
- Starts with EPR: if their reasoning is sound, then a deeper theory should exist with pre-existing properties (hidden variables) and no faster-than-light influence (locality).
- Bell formalized his notion of "local causality" as a mathematical condition on joint probabilities in any local hidden-variable theory (expressed as factorization).
  - Bell warned against taking his principle too seriously.
- Bell's theorem states that no theory with pre-existing variables and local causality can reproduce all quantum predictions.
- Bell's theorem has no bearing on the concept of "realism" as understood in the field of epistemology, and it does not rule out hidden-variable theories altogether.
  - Bell himself was strongly sympathetic to pilot-wave theory, a nonlocal hidden-variable theory.

GHZ Thought Experiment

- Consider three entangled particles distributed to Alice, Bob, and Charlie. Each can choose to measure either X or Y. Each individual outcome is ± 1 and appears random, but the joint results are correlated in the GHZ state.

Quantum Predictions:

- If they all measure X, their results multiply to -1.
- If one measures X and the other two measure Y, their results multiply to $+1$.



Local Hidden Variables

Classical Assumptions

- No causal influence can travel faster than light.
- Each particle has a fixed value before measurement (a hidden variable, not described by the wave function).

After becoming entangled, let the particles carry with them sets of instructions. The outcomes are predetermined before traveling to distant observers for measurement.

Hence, the values of $(A_X, A_Y), (B_X, B_Y), (C_X, C_Y)$ are fixed (equal to ± 1 each) and independent of even spacelike separated events (like distant measurements).

$$A_X B_Y C_Y = +1$$
$$A_Y B_X C_Y = +1$$
$$A_Y B_Y C_X = +1$$

} must be true

$$(A_X B_Y C_Y)(A_Y B_X C_Y)(A_Y B_Y C_X) = +1$$

$$A_X B_X C_X \, A_Y^2 B_Y^2 C_Y^2 = +1 \qquad (\pm 1)^2 = 1$$

$$A_X B_X C_X = +1$$ contradiction!

(i) ~~quantum predictions are wrong~~
(ii) signals are traveling faster than light ?
(iii) particles do not have fixed values before X/Y measurements ?

Example

$A_X = 1$	$B_Y = 1$	$C_Y = 1$	1
$A_Y = -1$	$B_X = -1$	$C_Y = 1$	1
$A_Y = -1$	$B_Y = 1$	$C_X = -1$	1
$A_X = 1$	$B_X = -1$	$C_X = -1$	1

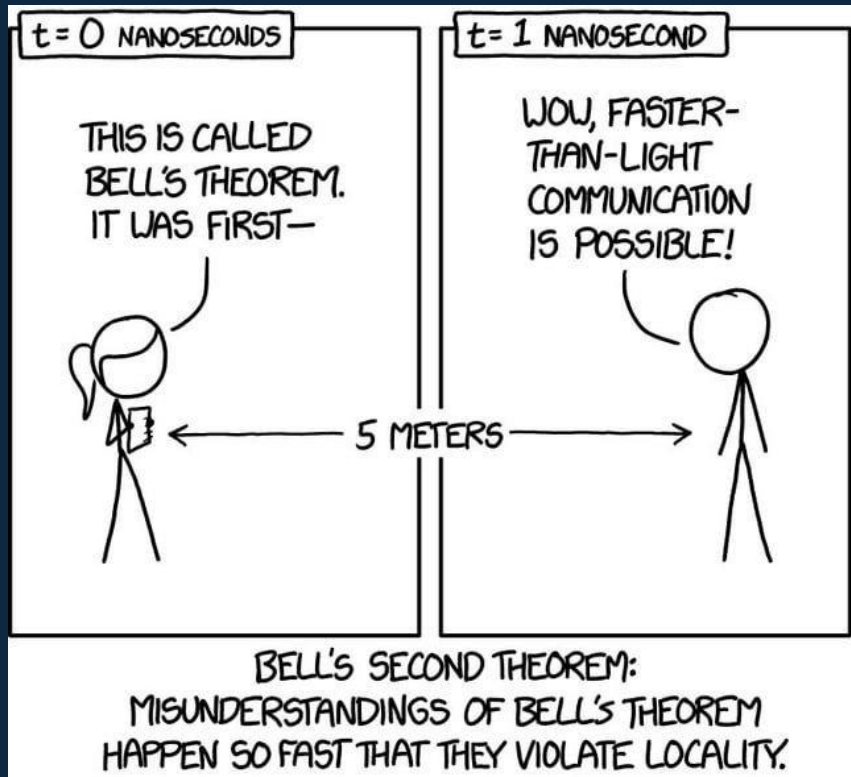
Try to arrange the values of $(A_X, A_Y, B_X, B_Y, C_X, C_Y)$ such that the first three rows multiply to $+1$ and the last row multiplies to -1. As we demonstrated on the last slide, it's not possible!

No-Signaling Theorem

Entanglement cannot be used to transmit information faster than light!

Even though entangled particles exhibit strong nonlocal correlations, no local operation on one particle can influence the measurement outcome statistics of another in a way that could be used for communication.

The “no-signaling theorem” does not imply that QM is a local theory.



Bell Inequalities

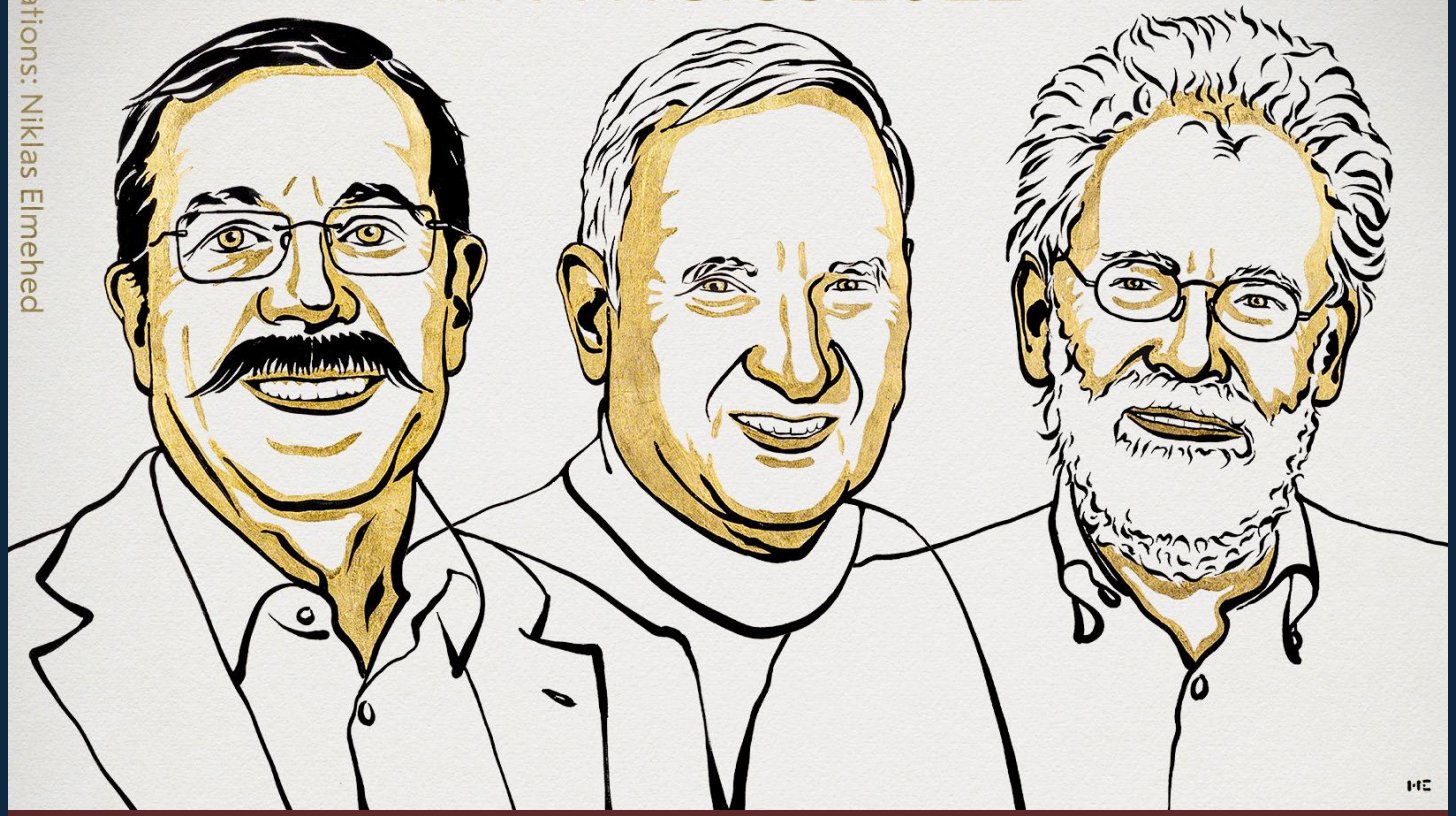
- Bell inequalities are constraints on correlations that follow from a set of assumptions about how physical systems behave (pre-existing variables and local causality)
- Quantum mechanics predicts violations of these inequalities, and the violations have been experimentally verified.
- Bell inequalities function as tests:
  - If the inequality holds, data could come from a local hidden-variable model.
  - If it is violated, at least one assumption (pre-existing values or local causality) must fail.
- Violations reveal “nonclassical correlations” that can be harnessed as a resource in quantum technologies.

Modern Bell Tests

- Early experiments (Aspect, 1980s) used entangled photons and already showed clear violations.
- Modern Bell tests go further by addressing potential “loopholes”
  - Detection loophole
  - Locality loophole
  - Freedom-of-choice loophole
- Since 2015, “loophole-free Bell tests” have been achieved.
  - Nitrogen-vacancy centers
  - Entangled photons at spacelike separation
  - High-efficiency photonic systems
- Across all these platforms, the observed correlations violate Bell inequalities and agree with the predictions of quantum mechanics.

THE NOBEL PRIZE IN PHYSICS 2022

Illustrations: Niklas Elmehed



Alain Aspect

John F. Clauser

Anton Zeilinger

"for experiments with entangled photons, establishing the violation of Bell inequalities and pioneering quantum information science"

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“Long may Louis de Broglie continue to inspire those who suspect that what is proved by impossibility proofs is lack of imagination.”

- John Bell